

Vanguard 16Ch Breast Tabletop Coil for GE 3T MR750 and MR750w Installation and Service Manual

Manufactured By:



Sentinelle Medical, a Division of Hologic

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Revision History

Rev	Date	Author	Type (Major/Minor)	Summary of Changes
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To file a report with GE:

Call 1-800-548-3366 and use option 6.

Fill out a report on <http://egems.med.ge.com/edq/home.jsp>

Contact your local service coordinator for more information on this process.

To file a report with Sentinelle:

Call 1-866-735-3744 and use option 1.

ATTENTION!

A GE field engineer or a Sentinelle Medical Field Service Technician **MUST** perform the installation of this coil. Please contact the field engineer from the company which sold the product to arrange for installation.

Attempts to perform installation of this coil without proper system configuration will result in system errors.

ПРЕДУПРЕЖДЕНИЕ (BG)	Това упътване за работа е налично само на английски език. - Ако доставчикът на услугата на клиента изиска друг език, задължение на клиента е да осигури превод. - Не използвайте оборудването, преди да сте се консултирали и разбрали упътването за работа. - Неспазването на това предупреждение може да доведе до нараняване на доставчика на услугата, оператора или пациента в резултат на токов удар, механична или друга опасност.
警告 (ZH-CN)	本维修手册仅提供英文版本。 - 如果客户的维修服务人员需要非英文版本，则客户需自行提供翻译服务。 - 未详细阅读和完全理解本维修手册之前，不得进行维修。 - 忽略本警告可能对维修服务人员、操作人员或患者造成电击、机械伤害或其他形式的伤害。
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UPOZORENJE (HR)	Ovaj servisni priručnik dostupan je na engleskom jeziku. - Ako davatelj usluge klijenta treba neki drugi jezik, klijent je dužan osigurati prijevod. - Ne pokušavajte servisirati opremu ako niste u potpunosti pročitali i razumjeli ovaj servisni priručnik. - Zanemarite li ovo upozorenje, može doći do ozljede davatelja usluge, operatera ili pacijenta uslijed strujnog udara, mehaničkih ili drugih rizika.
VÝSTRAHA (CS)	Tento provozní návod existuje pouze v anglickém jazyce. - V případě, že externí služba zákazníkům potřebuje návod v jiném jazyce, je zajištění překladu do odpovídajícího jazyka úkolem zákazníka. - Nesnažte se o údržbu tohoto zařízení, aniž byste si přečetli tento provozní návod a pochopili jeho obsah. - V případě nedodržování této výstrahy může dojít k poranění pracovníka prodejního servisu, obslužného personálu nebo pacientů vlivem elektrického proudu, respektive vlivem mechanických či jiných rizik.
ADVARSEL (DA)	Denne servicemanual findes kun på engelsk. - Hvis en kundes tekniker har brug for et andet sprog end engelsk, er det kundens ansvar at sørge for oversættelse. - Forsøg ikke at servicere udstyret uden at læse og forstå denne servicemanual. - Manglende overholdelse af denne advarsel kan medføre skade på grund af elektrisk stød, mekanisk eller anden fare for tekniker, operatøren eller patienten.
WAARSCHUWING (NL)	Deze onderhoudshandleiding is enkel in het Engels verkrijgbaar. - Als het onderhoudspersoneel een andere taal vereist, dan is de klant verantwoordelijk voor de vertaling ervan. - Probeer de apparatuur niet te onderhouden alvorens deze onderhoudshandleiding werd geraadpleegd en begrepen is. - Indien deze waarschuwing niet wordt opgevolgd, zou het onderhoudspersoneel, de operator of een patiënt gewond kunnen raken als gevolg van een elektrische schok, mechanische of andere gevaren.
WARNING (EN)	This service manual is available in English only. - If a customer's service provider requires a language other than english, it is the customer's responsibility to provide translation services. - Do not attempt to service the equipment unless this service manual has been consulted and is understood. - Failure to heed this warning may result in injury to the service provider, operator or patient from electric shock, mechanical or other hazards.
HOIATUS (ET)	See teenindusjuhend on saadaval ainult inglise keeles - Kui klienditeeninduse osutaja nõuab juhendit inglise keelest erinevas keeles, vastutab klient tõlketeenuse osutamise eest. - Ärge üritage seadmeid teenindada enne eelnevalt käesoleva teenindusjuhendiga tutvumist ja sellest aru saamist. - Käesoleva hoiatuse eiramine võib põhjustada teenuseosutaja, operaatori või patsiendi vigastamist elektrilöögi, mehaanilise või muu ohu tagajärjel.
VAROITUS (FI)	Tämä huolto-ohje on saatavilla vain englanniksi. - Jos asiakkaan huoltohenkilöstö vaatii muuta kuin englanninkielistä materiaalia, tarvittavan käännöksen hankkiminen on asiakkaan vastuulla. - Älä yritä korjata laitteistoa ennen kuin olet varmasti lukenut ja ymmärtänyt tämän huolto-ohjeen. - Mikäli tätä varoitusta ei noudateta, seurauksena voi olla huoltohenkilöstön, laitteiston käyttäjän tai potilaan vahingoittuminen sähköiskun, mekaanisen vian tai muun vaaratilanteen vuoksi.
ATTENTION (FR)	Ce manuel d'installation et de maintenance est disponible uniquement en anglais. - Si le technicien d'un client a besoin de ce manuel dans une langue autre que l'anglais, il incombe au client de le faire traduire. - Ne pas tenter d'intervenir sur les équipements tant que ce manuel d'installation et de maintenance n'a pas été consulté et compris. - Le non-respect de cet avertissement peut entraîner chez le technicien, l'opérateur ou le patient des blessures dues à des dangers électriques, mécaniques ou autres.

WARNUNG (DE)	Diese Serviceanleitung existiert nur in englischer Sprache. - Falls ein fremder Kundendienst eine andere Sprache benötigt, ist es Aufgabe des Kunden für eine entsprechende Übersetzung zu sorgen. - Versuchen Sie nicht diese Anlage zu warten, ohne diese Serviceanleitung gelesen und verstanden zu haben. - Wird diese Warnung nicht beachtet, so kann es zu Verletzungen des Kundendiensttechnikers, des Bedieners oder des Patienten durch Stromschläge, mechanische oder sonstige Gefahren kommen.
ΠΡΟΕΙΔΟΠΟΙΗΣΗ (EL)	Το παρόν εγχειρίδιο σέρβις διατίθεται μόνο στα αγγλικά. - Εάν ο τεχνικός σέρβις ενός πελάτη απαιτεί το παρόν εγχειρίδιο σε γλώσσα εκτός των αγγλικών, αποτελεί ευθύνη του πελάτη να παρέχει τις υπηρεσίες μετάφρασης. - Μην επιχειρήσετε την εκτέλεση εργασιών σέρβις στον εξοπλισμό αν δεν έχετε συμβουλευτεί και κατανοήσει το παρόν εγχειρίδιο σέρβις. - Αν δεν προσέξετε την προειδοποίηση αυτή, ενδέχεται να προκληθεί τραυματισμός στον τεχνικό σέρβις, στο χειριστή ή στον ασθενή από ηλεκτροπληξία, μηχανικούς ή άλλους κινδύνους.
FIGYELMEZTETÉS (HU)	Ezen karbantartási kézikönyv kizárólag angol nyelven érhető el. - Ha a vevő szolgáltatója angoltól eltérő nyelvre tart igényt, akkor a vevő felelőssége a fordítás elkészítése. - Ne próbálja elkezdni használni a berendezést, amíg a karbantartási kézikönyvben leírtakat nem értelmezték. - Ezen figyelmeztetés figyelmen kívül hagyása a szolgáltató, működtető vagy a beteg áramütés, mechanikai vagy egyéb veszélyhelyzet miatti sérülését eredményezheti.
ADVÖRUN (IS)	Þessi þjónustuhandbók er aðeins fáanleg á ensku. - Ef að þjónustuveitandi viðskiptamanns þarfnast annas tungumáls en ensku, er það skylda viðskiptamanns að skaffa tungumálaþjónustu. - Reynið ekki að afgreiða tækið nema að þessi þjónustuhandbók hefur verið skoðuð og skilið. - Brot á sinna þessari aðvörun getur leitt til meiðsla á þjónustuveitanda, stjórnanda eða sjúklings frá raflosti, vélrænu eða öðrum áhættum.
AVVERTENZA (IT)	Il presente manuale di manutenzione è disponibile soltanto in lingua inglese. - Se un addetto alla manutenzione richiede il manuale in una lingua diversa, il cliente è tenuto a provvedere direttamente alla traduzione. - Procedere alla manutenzione dell'apparecchiatura solo dopo aver consultato il presente manuale ed averne compreso il contenuto. - Il mancato rispetto della presente avvertenza potrebbe causare lesioni all'addetto alla manutenzione, all'operatore o ai pazienti provocate da scosse elettriche, urti meccanici o altri rischi.
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경고 (KO)	본 서비스 매뉴얼은 영어로만 이용하실 수 있습니다. 고객의 서비스 제공자가 영어 이외의 언어를 요구할 경우, 번역 서비스를 제공하는 것은 고객 책임입니다. 본 서비스 매뉴얼을 참조하여 숙지하지 않은 이상 해당 장비를 수리하려고 시도하지 마십시오. 본 경고 사항에 유의하지 않으면 전기 쇼크, 기계적 위험, 또는 기타 위험으로 인해 서비스 제공자, 사용자 또는 환자에게 부상을 입힐 수 있습니다.
BRĪDINĀJUMS (LV)	Šī apkopes rokasgrāmata ir pieejama tikai angļu valodā. - Ja klienta apkopes sniedzējam nepieciešama informācija citā valodā, klienta pienākums ir nodrošināt tulkojumu. - Neveiciet aprikojuma apkopi bez apkopes rokasgrāmatas izlasīšanas un saprašanas. - Šī brīdinājuma neievērošanas rezultātā var rasties elektriskās strāvas trieciena, mehānisku vai citu faktoru izraisītu traumu risks apkopes sniedzējam, operatoram vai pacientam.
ISPĖJIMAS (LT)	Šis eksploatavimo vadovas yra tik anglų kalba. - Jei kliento paslaugų tiekėjas reikalauja vadovo kita kalba – ne anglų, suteikti vertimo paslaugas privalo klientas. - Nemėginkite atlikti įrangos techninės priežiūros, jei neperskaitėte ar nesupratote šio eksploatavimo vadovo. - Jei nepaisysite šio įspėjimo, galimi paslaugų tiekėjo, operatoriaus ar paciento sužalojimai dėl elektros šoko, mechaninių ar kitų pavojų.
ADVARSEL (NO)	Denne servicehåndboken finnes bare på engelsk. - Hvis kundens serviceleverandør har bruk for et annet språk, er det kundens ansvar å sørge for oversettelse. - Ikke forsøk å reparere utstyret uten at denne servicehåndboken er lest og forstått. - Manglende hensyn til denne advarselen kan føre til at serviceleverandøren, operatøren eller pasienten skades på grunn av elektrisk støt, mekaniske eller andre farer.
OSTRZEŻENIE (PL)	Niniejszy podręcznik serwisowy dostępny jest jedynie w języku angielskim. - Jeśli serwisant klienta wymaga języka innego niż angielski, zapewnienie usługi tłumaczenia jest obowiązkiem klienta. - Nie próbować serwisować urządzenia bez zapoznania się z niniejszym podręcznikiem serwisowym i zrozumienia go. - Niezastosowanie się do tego ostrzeżenia może doprowadzić do obrażeń serwisanta, operatora lub pacjenta w wyniku porażenia prądem elektrycznym, zagrożenia mechanicznego bądź innego.
AVISO (PT-BR)	Este manual de assistência técnica encontra-se disponível unicamente em inglês. - Se outro serviço de assistência técnica solicitar a tradução deste manual, caberá ao cliente fornecer os serviços de tradução. - Não tente reparar o equipamento sem ter consultado e compreendido este manual de assistência técnica. - A não observância deste aviso pode ocasionar ferimentos no técnico, operador ou paciente decorrentes de choques elétricos, mecânicos ou outros.

ATENÇÃO (PT-PT)	Este manual de assistência técnica só se encontra disponível em inglês. - Se qualquer outro serviço de assistência técnica solicitar este manual noutro idioma, é da responsabilidade do cliente fornecer os serviços de tradução. - Não tente reparar o equipamento sem ter consultado e compreendido este manual de assistência técnica. - O não cumprimento deste aviso pode colocar em perigo a segurança do técnico, do operador ou do paciente devido a choques eléctricos, mecânicos ou outros.
ATENȚIE (RO)	Acest manual de service este disponibil doar în limba engleză. - Dacă un furnizor de servicii pentru clienți necesită o altă limbă decât cea engleză, este de datoria clientului să furnizeze o traducere. - Nu încercați să reparați echipamentul decât ulterior consultării și înțelegerii acestui manual de service. - Ignorarea acestui avertisment ar putea duce la rănirea depanatorului, operatorului sau pacientului în urma pericolelor de electrocutare, mecanice sau de altă natură.
ОСТОРОЖНО! (RU)	Данное руководство по техническому обслуживанию представлено только на английском языке. - Если сервисному персоналу клиента необходимо руководство не на английском, а на каком-то другом языке, клиенту следует самостоятельно обеспечить перевод. - Перед техническим обслуживанием оборудования обязательно обратитесь к данному руководству и поймите изложенные в нем сведения. - Несоблюдение требований данного предупреждения может привести к тому, что специалист по техобслуживанию, оператор или пациент получит удар электрическим током, механическую травму или другое повреждение.
UPOZORENJE (SR)	Ovo servisno uputstvo je dostupno samo na engleskom jeziku. - Ako klijentov serviser zahteva neki drugi jezik, klijent je dužan da obezbedi prevodilačke usluge. - Ne pokušavajte da opravite uređaj ako niste pročitali i razumeli ovo servisno uputstvo. - Zanemarivanje ovog upozorenja može dovesti do povređivanja serviser, rukovaoca ili pacijenta usled strujnog udara ili mehaničkih i drugih opasnosti.
UPOZORNENIE (SK)	Tento návod na obsluhu je k dispozícii len v angličtine. - Ak zákaznikov poskytovateľ služieb vyžaduje iný jazyk ako angličtinu, poskytnutie prekladateľských služieb je zodpovednosťou zákazníka. - Nepokúšajte sa o obsluhu zariadenia, kým si neprečítate návod na obsluhu a neporozumiete mu. - Zanedbanie tohto upozornenia môže spôsobiť zranenie poskytovateľa služieb, obsluhujúcej osoby alebo pacienta elektrickým prúdom, mechanické alebo iné ohrozenie.
ATENCION (ES)	Este manual de servicio sólo existe en inglés. - Si el encargado de mantenimiento de un cliente necesita un idioma que no sea el inglés, el cliente deberá encargarse de la traducción del manual. - No se deberá dar servicio técnico al equipo, sin haber consultado y comprendido este manual de servicio. - La no observancia del presente aviso puede dar lugar a que el proveedor de servicios, el operador o el paciente sufran lesiones provocadas por causas eléctricas, mecánicas o de otra naturaleza.
VARNING (SV)	Den här servicehandboken finns bara tillgänglig på engelska. - Om en kunds servicetekniker har behov av ett annat språk än engelska, ansvarar kunden för att tillhandahålla översättningstjänster. - Försök inte utföra service på utrustningen om du inte har läst och förstår den här servicehandboken. - Om du inte tar hänsyn till den här varningen kan det resultera i skador på serviceteknikern, operatören eller patienten till följd av elektriska stötar, mekaniska faror eller andra faror.
OPOZORILO (SL)	Ta servisni priročnik je na voljo samo v angleškem jeziku. - Če ponudnik storitve stranke potrebuje priročnik v drugem jeziku, mora stranka zagotoviti prevod. - Ne poskušajte servisirati opreme, če tega priročnika niste v celoti prebrali in razumeli. - Če tega opozorila ne upoštevate, se lahko zaradi električnega udara, mehanskih ali drugih nevarnosti poškoduje ponudnik storitev, operater ali bolnik.
DİKKAT (TR)	Bu servis kılavuzunun sadece ingilizcesi mevcuttur. - Eğer müşteri teknisyeni bu kılavuzu ingilizce dışında bir başka lisandan talep ederse, bunu tercüme ettirmek müşteriye düşer. - Servis kılavuzunu okuyup anlamadan ekipmanlara müdahale etmeyiniz. - Bu uyarıya uyulmaması, elektrik, mekanik veya diğer tehlikelerden dolayı teknisyen, operatör veya hastanın yaralanmasına yol açabilir.

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Note:

There are sections in this manual that apply only to Sentinelle Medical (SMI) personnel or GE Healthcare (GEHC) personnel. These sections are noted using the following symbols.



OR



GE FE's shall only service systems that were sold by GEHC. Coils sold by GEHC will have a "Distributed by GE Healthcare" label next to the rating plate on the underside of the patient support.

1 Purpose

The purpose of this manual is to provide detailed instructions to an SMI Service Technician or a GEHC field engineer for installation, testing, service and maintenance of the Vanguard 16ch Tabletop for GE MR750 and MR750w 3T MRI scanners. The Troubleshooting Table outlines the required steps to methodically assess and correct the reported problem(s).

Information reported from the site contact explaining the problem is paramount and the Service Technician must record the details. The Service Technician must carefully gather and interpret the information received from the site contact, to ensure all details are enclosed to enable the problem to be quickly diagnosed and repaired.

2 Compatibility

This device is compatible with Discovery MR750 and MR750w 3T MRI Scanner operating on '22.0' or later software.

3 Service Tools Required

Many of these tools are included in the standard toolkit.

Item	GEHC Part #	SMI Part #	Qty	Description
1	5109895-13	926-3A1-07B	1	Slot screwdriver medium
2	5109892-6	03-L-4	1	Allen Wrench 1/8"
3	5109892-3	Part of 1-927-16537	1	Allen Wrench 5/64"

3.1 General Safety Requirements



Follow all required safety and PPE procedures customary for your organization when working on this product.

- Wear safety shoes for foot protection.
- Wear gloves for hand protection.
- Wear safety glasses for eye protection.

3.1.1 Occupational Exposure to Infectious Agents

Installing and servicing the Vanguard requires visiting medical facilities which present infection hazards, especially from possible exposure to pathogens found in blood and other potentially infectious bodily materials. Pathogens are microorganisms that can cause disease when transmitted from an infected individual to another individual through blood and certain other body materials. The most common illnesses caused by blood borne pathogens are hepatitis B (HBV), hepatitis C (HCV), and acquired immunodeficiency syndrome (AIDS) from HIV, human immunodeficiency virus. These blood borne pathogens are capable of causing serious illness and death.



Failure to adhere to these procedures and precautions may result in infection, serious illness and death.

Types of Hazardous Exposure

Hazardous exposures which may occur in medical facilities include:

- Percutaneous inoculations or punctures with blood or body fluid by a sharp instrument or needle.
- Contact with blood or body fluid through a fresh cut (less than 24 hours from occurrence) or mucous membrane contact (e.g., a splash to the eye or mouth, or mouth-to-mouth resuscitation).
- Skin exposure involving large amounts of blood or prolonged contact with blood, especially when the exposed skin is chapped, abraded, or afflicted with dermatitis.

Avoiding Exposure

Before beginning work at a medical facility where you may encounter potential infection hazards:

- Ensure that you have had a Hepatitis B vaccination.
- Before beginning to work on equipment, check carefully for signs of blood or body fluids.
- If you notice possible blood or body fluids on the equipment, report it to the responsible facility personnel and have them clean the equipment using their institutional infection control procedures.
- If there is a reasonable expectation of the presence of infectious materials, do not work on the equipment until it is disinfected.
- As a precautionary measure, wear personal protective equipment including gloves, protective eye covering and disposable clothing whenever the potential to be exposed to blood or other potentially infected materials exist.

After Exposure

If you contact a potentially infectious blood or body fluid:

1. Seek immediate medical assistance.
2. Advise your supervisor of the exposure event.

GEHC ONLY

- If you have previously refused vaccination to Hepatitis B, GE must offer the vaccination to you at this time at no charge.
- As part of the investigation process, contact the GE medical team to assist in identifying the status of source individuals.

3. Document the incident for your employer and the facility where the incident occurred:
 - o If feasible, document the name and address of the individual source of the possible infection.
 - o Document the time, place, and circumstances of the contact.

4 Definitions and Abbreviations

Throughout this document the following terms and abbreviations will be used:

FRU	Field Replaceable Unit
Installation Verification Checklist	SMI-1487 Vanguard for MR750 and MR750w Installation Verification
Left/Right	Directions relative to the patient lying in prone position
MCQA	Multi-Coil Quality Assurance
Patient Support	Mechanical structure supporting the patient
Service Toolkit	SMI-0294 Toolkit utilized by SMI Field Service Technician
SMI	Sentinelle Medical
User Manual	SMI-1020 GE Tabletop User Manual

5 Performing Service

5.1 Precautions

Before entering the MRI Room with the Vanguard system the following activities must be performed:

- Thoroughly inspect the system to ensure loose ferrous/magnetic items are not present anywhere on the system. Loose objects can become airborne and/or travel at high velocity in the presence of the scanner, potentially causing harm to the Patient, Operator or equipment.

SMI ONLY	• Do not attempt to make any changes or adjustments to the GE scanner or accessories. If any damage to the GE scanner is observed, report to the customer. • Do not touch metal pins of any connectors. Leave the protective packaging on all connectors until install is complete to prevent electrostatic discharge damage. • Patient data is confidential – be sure not to examine, copy or photograph any printed or electronic matter containing patient names, personal details, medical images or other confidential data to ensure compliance with the Health Insurance Portability and Accountability Act (HIPPA).
----------	---

5.2 | | | |----------|--------------------------------| | SMI ONLY | Preparing for the Service Call | |----------|--------------------------------|

Prior to performing the service call the following are required:

- Installation should be booked before application training is scheduled to start. Follow up 3 days before the installation date to remind the customer and confirm that the scanner time is booked off and the Vanguard has been delivered to the immediate area of the scanner.
- A copy of the P.O. or work order for the unit to be installed.
- Contact information for the site (usually the charge tech or administrator) and back-up contact person.
- Determine if any nearby sites can be visited during the same trip to perform preventative maintenance.
- Bring a complete service kit, camera (if possible), notebook, service report cards and personal business cards.
- When international travel is required, a signed letter from the service manager or other manager explaining the purpose of the trip, the destination, dates and the employment status of the service technician.
- ID badge, to be kept on person at all times, and passport/visa (as needed).
- Hotel info, customer site address, directions to/from site, airport and hotel.
- If possible, contact the site specific GE Service Personnel to inform them of the pending service call and request information on any recent problems with or service to the scanner (SMI service calls can be generated from recent service to the scanner involving adjustment of mechanical parts or software upgrades and patches). Request they be present to provide the service key (code).
- Ensure that only Sentinelle-authorized service personnel make adjustments or repairs to the system.
- Review Sentinelle service logs for the site for previous service reports and service pending.

5.3 Recording the Service Call

Record the results of all activities performed including:

- Any non-conformances that were discovered during unpackaging and installation or service and any remedial action taken.
- Leave a completed service card and inform relevant staff members who are present that service is complete and indicate state of functionality. If the system fails any of the functionality tests in the installation procedure, it must be taken out of service.
- Unless prior arrangements have been made with the Applications Support Specialists, the system will normally be “Out of Service – Pending Applications training” immediately following installation.
- An SMI or affiliate service log must be filled out to officially record the installation.
- If any non-conformances or problems are discovered during installation, it is imperative that these are recorded, reported and entered into the Problem Management / Complaint Database.

5.4 Departure

- Ensure work area is clean and no tools or garbage are left behind.
- Replace any coils and padding previously on the scanner bed.
- Place all Vanguard components and accessories in a common storage area.

SMI
ONLY

- Complete a Sentinelle calling card and leave it in a prominent place, preferably at the console.
- If possible, inform staff member on site of completion before leaving.
- Ensure doors (especially scanner room door) are locked and lights are off in accordance with the customer site's requirements.

6 System Contents

This section outlines the shipping contents of this system and locates the rating plate labels for product identification.

6.1 Shipping Contents

The Vanguard components are shipped in three groups; the Patient Support and accessories, RF system and padding kit. All should be verified with included packing list. Refer to page 3 of this manual for the **Damage in Transportation** section if damage is found.

6.1.1 Patient Support and Accessories

Item	Box #	GEHC Part #	SMI Part #	Description
1	2 of 2	5408232	4000189-11	Patient Support Assembly
2	2 of 2	5408235	SVC-00319	Shoulder Bridge (Qty 2)
3	2 of 2	5408236	SVC-00318	Headrest Assembly
4	2 of 2	5408240	4000210-11	Compression Slider Left
5	2 of 2	5408241	4000290-11	Compression Slider Right
6	2 of 2	5408245	4000196-11	Compression Frame Ass'y Left
7	2 of 2	5408244	4000197-11	Compression Frame Ass'y Right
8	2 of 2	5408233	SVC-00330	Contralateral Support
9	2 of 2	5408234	SVC-00331	Catchment
10	2 of 2	5408286	4001129-51	Asset Pad, Lateral (Qty 2)
11	2 of 2	5408286	4001129-51	Asset Pad, Medial
12	2 of 2	5407765-11	SVC-00487	GE Spherical Phantom Positioner
14	2 of 2	E8811EB	46-25845G1	Kit/Strap (not shown)

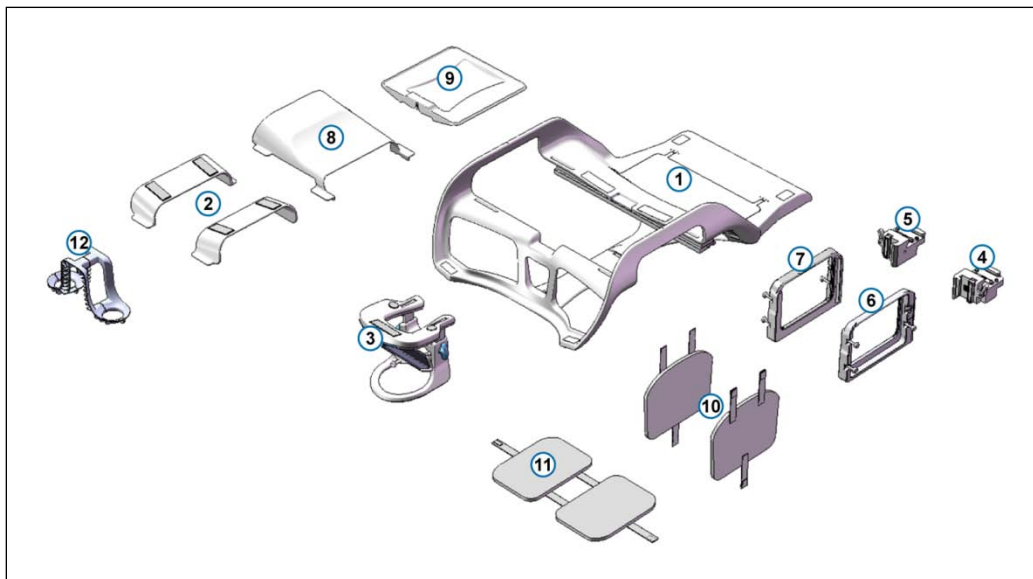


Figure 1

6.1.2 RF System

Item	Box #	GEHC Part #	SMI Part #	Description
1	1 of 2	5407765-8	4000444-11	Interface Box 3T16Ch
2	1 of 2	5407765-6	4000445-11	Medial Array 3T16Ch
3	1 of 2	5407765-4	4000429-11	Lateral Array Right 3T16Ch
4	1 of 2	5407765-5	4000428-11	Lateral Array Left 3T16Ch
5	1 of 2	5407765-3	4000448-11	Single Loop Right 3T16Ch
6	1 of 2	5407765-2	4000452-11	Single Loop Left 3T16Ch
7	1 of 2	5407765-7	4000349-11	Medial Plug 3T16Ch

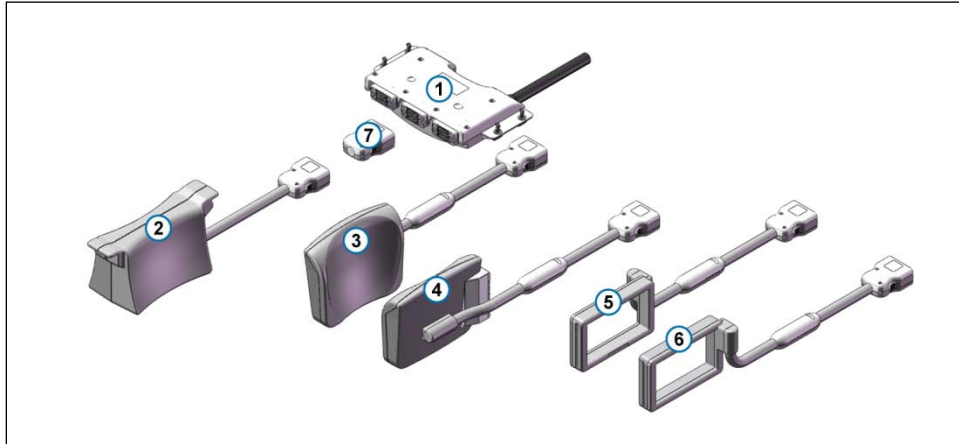


Figure 2

6.1.3 Padding Kit

Item	Box #	GEHC Part #	SMI Part #	Description
1	2 of 2	5408291	SVC-00325	Pad Arm Support
2	2 of 2	In 5408290 Kit	SVC-00326	Ramp Pad
3	2 of 2	In 5408290 Kit	SVC-00327	Shoulder Bridge Pad (Qty 2)
4	2 of 2	5408289	SVC-00328	Body Pad
5	2 of 2	5408288	SVC-00309	Wedge Pad
6	2 of 2	5408287	SVC-00308	Headrest Pad
7	2 of 2	In 5408290 Kit	SVC-00322	Medial Pad
8	2 of 2	In 5408290 Kit	SVC-00320	Foot Pad

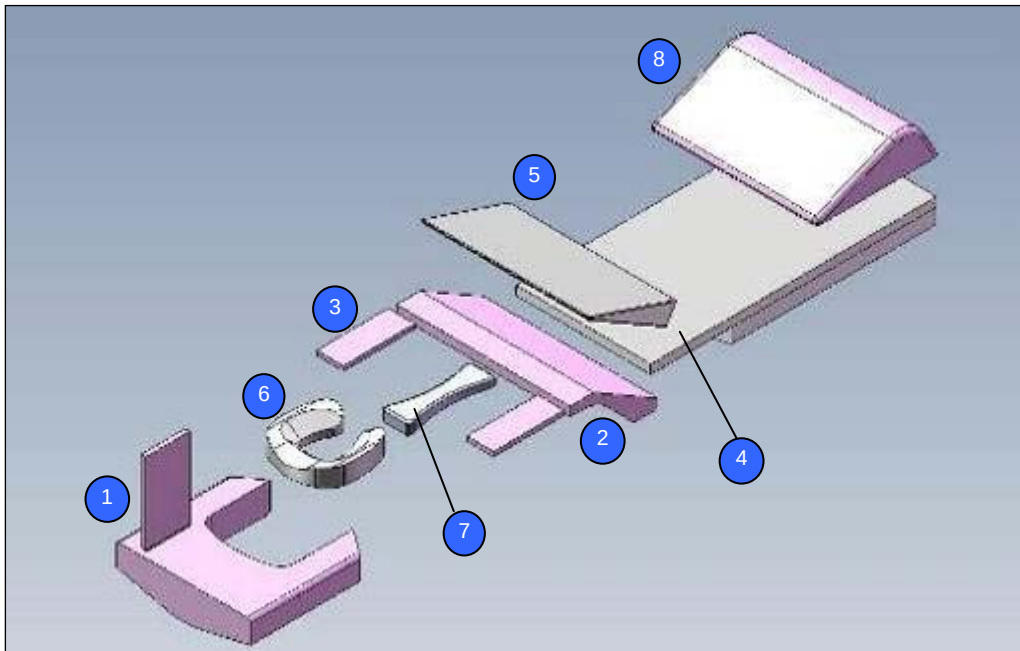


Figure 3

6.1.4 Miscellaneous Items

Item	Box #	GEHC Part #	SMI Part #	Description
1	2 of 2	5407766	SFW-01230	Coil CD
2	2 of 2	N/A	SMI-1414	Service Manual
3	2 of 2	N/A	SMI-1646	Packing List

6.1.5 Riser Kit for GE Discovery 750w Systems with GEM Option

Item	Box #	GEHC Part #	SMI Part #	Description
1	1 of 1	5407765-10	ASY-04625	Riser
2	1 of 1	5407765-13	SVC-00486	Riser Body Pad
3	1 of 1	5407765-14	SVC-00485	Riser Front Pad

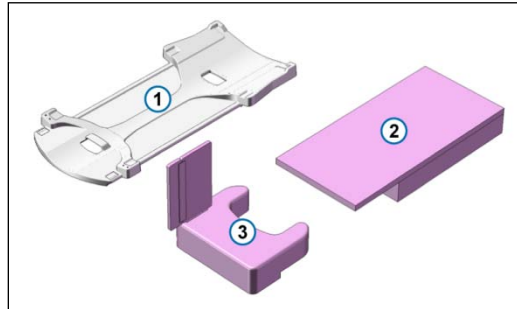


Figure 4

6.2 Product Identification

The system is uniquely identified using the labels below. For clarity, the labels are surrounded by a red box in the figures below. Systems sold and serviced by GEHC will have an additional “Distributed by: GE Healthcare” label next to the rating plate (Figure 4).

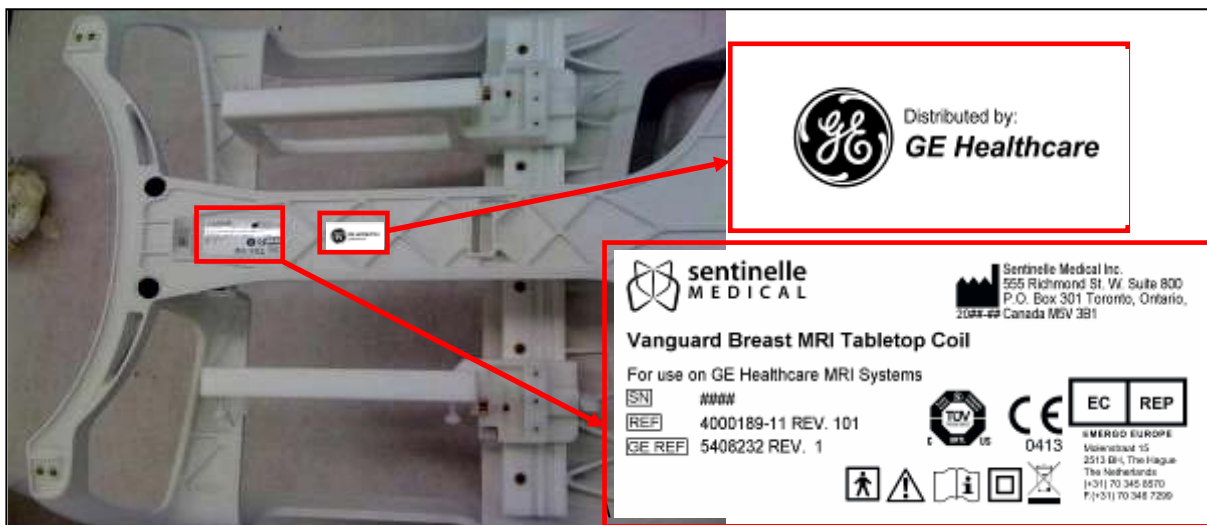


Figure 5: Patient Support Label



Figure 6: Interface Box Label

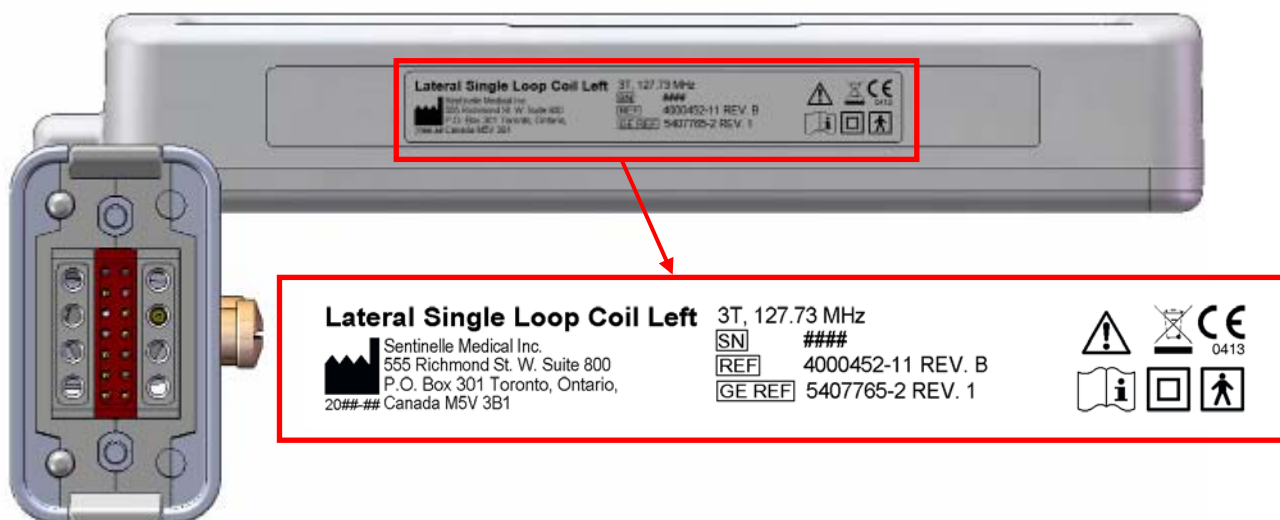


Figure 7: Lateral Single Loop Coil Left Label

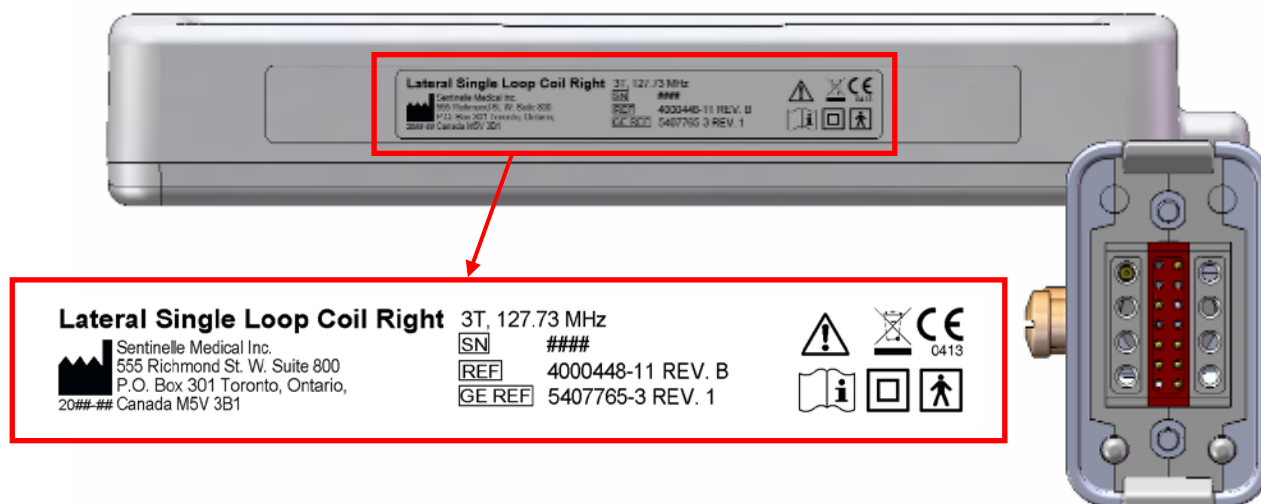


Figure 8: Lateral Single Loop Coil Right Label



Figure 9: Lateral Array Coil Left Label

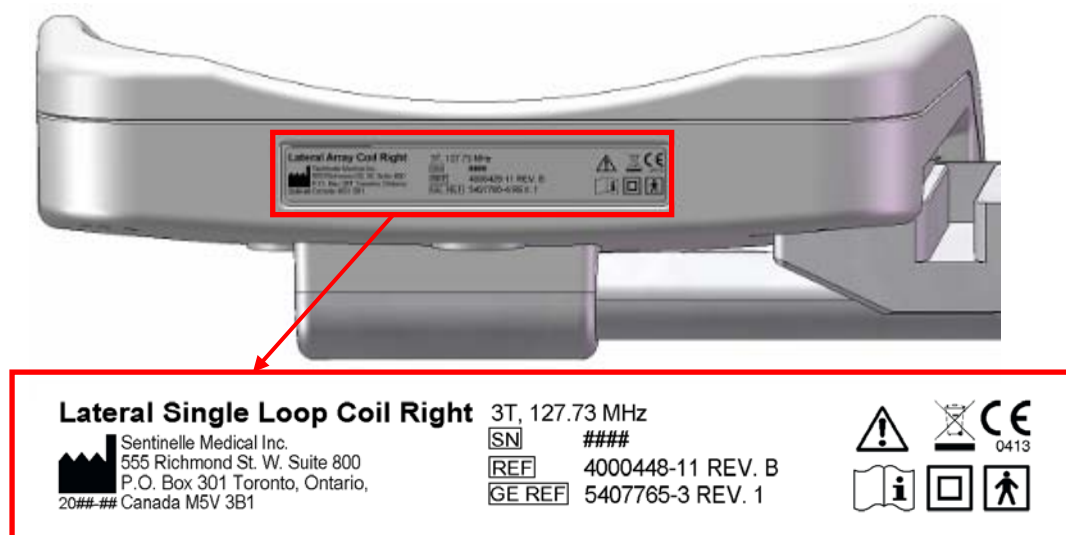


Figure 10: Lateral Array Coil Right Label

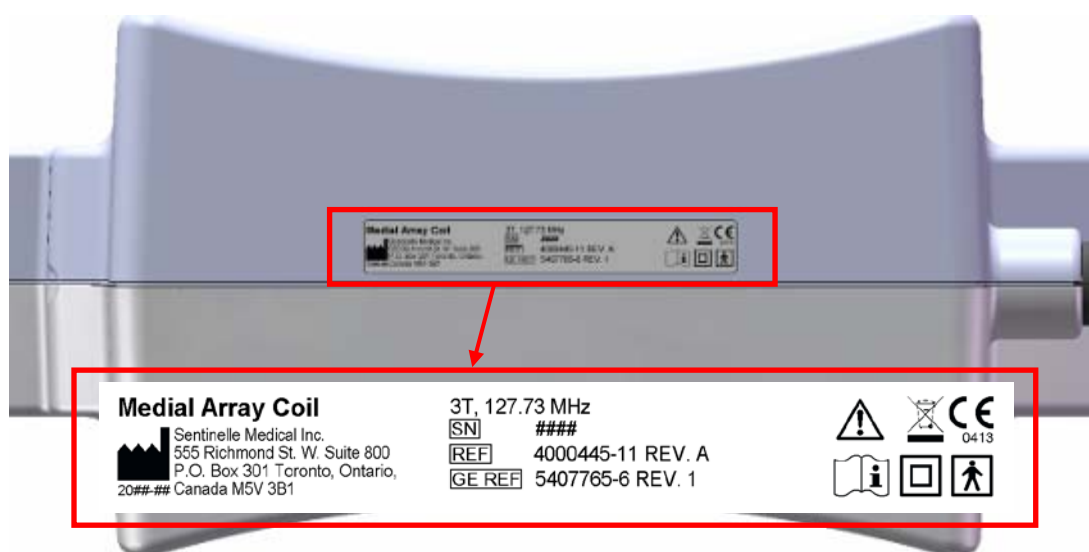


Figure 11: Medial Array Coil Label

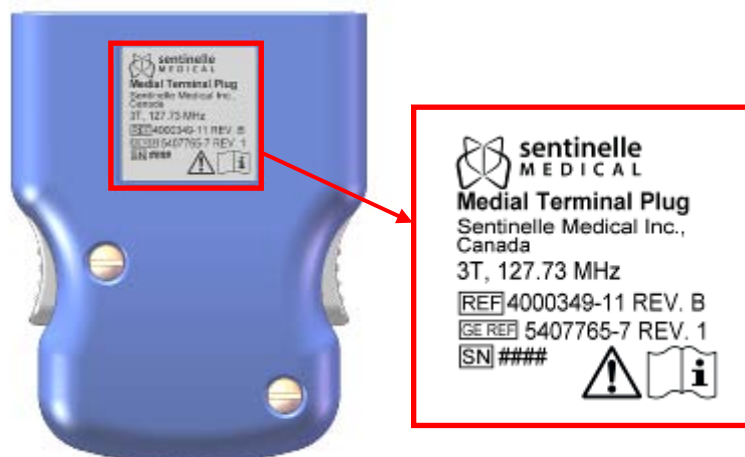


Figure 12: Medial Terminator Plug Label

7 Installation Procedure

Required installation time: 30-60 minutes.

Installation of the Vanguard requires no tools and no mechanical settings to be performed. The accessories must all be connected and assembled and tested to ensure system is functioning correctly with the scanner.

7.1 Unpack and Set Up

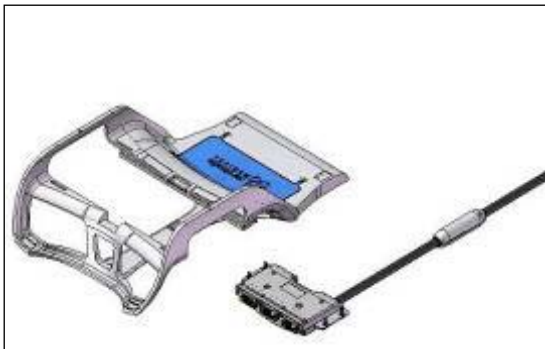
Unpack the Patient Support assembly, padding kit, riser kit (if applicable), and coil boxes while examining the components for damage. Verify all components and accessories are present including the User Manual. Store all components neatly inside the scanner room or as instructed by customer.

NOTE: Customers who have purchased the riser kit will have both the standard padding kit and the thicker riser body pad and riser front pad from the riser kit. You should segregate the standard body pad and front pad and inform the customer that they may either dispose of these pads or keep them in the event that they decide to use the Patient Support on a scanner with a curved patient table (GE HDx, GE Discovery 750 System, or GE Discovery 750w System without GEM Option).

*The riser must **always** be used with the thicker riser body pad and riser front pad.*

7.2 Interface Box Installation

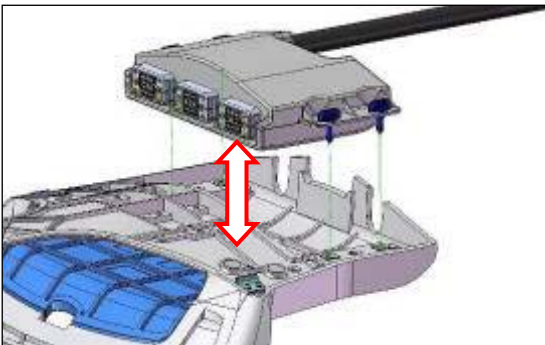
The interface box must be installed on the underside of the Patient Support. Follow the steps below.



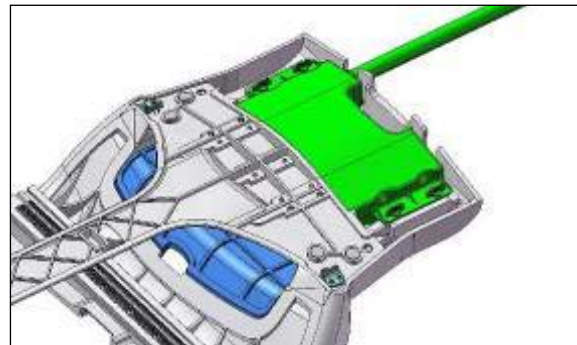
1. Prepare Patient Support and interface box.



2. Position Patient Support to access underside.



3. Align interface box to mounting holes.



4. Attach interface box to Patient Support with four fasteners.

7.3 Vanguard Installation

This section outlines the procedure that must be followed to setup the Patient Support and accessories for use. Verify all parts function correctly and are not damaged.

7.3.1 Patient Support Setup for the GE Discovery 750 System and GE Discovery 750w System without GEM Option

Place the Tabletop on the 3.0T GE MRI System patient table with the foot end toward the bore so the Tabletop cable located at the foot end of the patient support can be securely plugged into the MR scanner.



Figure 13 Patient Support on GE Patient Table



CAUTION

Do not use the riser for the GE Discovery 750w System with GEM Option on the GE Discovery 750 System or GE Discovery 750w System without GEM Option. Using the riser on these systems may damage the equipment and potentially injure patients.

7.3.1.1. Using the Anti-Tip Velcro straps

To prevent the patient support from tipping, the anti-tip Velcro straps must be properly attached before allowing a patient to mount the patient support. Insert a Velcro strap into the slot on each side of the 3.0T GE MRI System table, then slide and fasten them to the Tabletop.



Figure 14 Securing the Patient Support with the Anti-Tip Velcro Straps

7.3.2 Patient Support Setup for the GE Discovery 750w System with GEM Option



CAUTION

You must use the riser with the GE Discovery 750w System with GEM Option. Failure to use the riser may cause the Tabletop to tip over.

To set up the Tabletop on the GE Discovery 750w System with GEM Option:

1. Place the riser on the patient table with the rounded end of the riser pointing away from the bore.
2. Place the patient support on the riser with the head end of the patient support toward the rounded end of the riser.
3. Attach the patient support to the riser using the Velcro straps (circled in the figure below, left).
4. Secure the riser to the patient table by fastening the anti-tip Velcro straps to the Velcro pads at the head end of the riser. To provide additional stability, you can also fasten anti-tip Velcro straps to the Velcro on the reverse side of the straps that connect the patient support to the riser.

For more information about using the anti-tip Velcro straps, see Using the Anti-Tip Velcro Straps above.

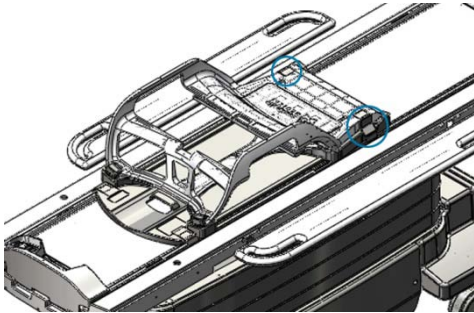


Figure 15 Patient Support Connected to the Riser

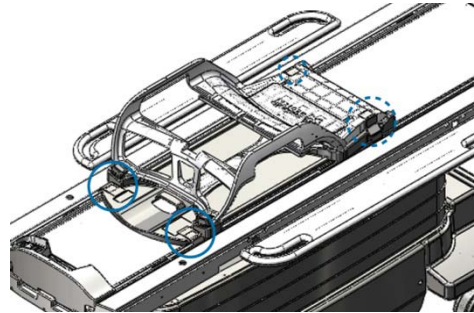


Figure 16 Velcro straps to Connect the Riser and Patient Support to the Patient Table

NOTE: If the Tabletop will be used on a GE Discovery 750w System with GEM Option only, you can leave the patient support connected to the riser and store them together.

7.3.3 General System Function and Components

The Tabletop is a dedicated breast imaging coil and interventional system for 3.0T GE MRI Systems with P Ports. The Tabletop consists of a patient support with breast immobilization system and a 2-, 10- or 16-channel coil system.

The patient support holds the patient comfortably in a prone position with the breasts lying pendant into the aperture.

The immobilization system for clinical imaging utilizes the lateral coils to immobilize the breast tissue. The immobilization for interventional procedures utilizes immobilization frames. Both the lateral coils and immobilization frames install in the sliders allowing for vertical and horizontal movement of the coils. The Tabletop's unique Variable Coil Geometry enables the coils to be positioned in close proximity to the breast tissue for improved signal-to-noise ratio (SNR).

For clinical imaging, coil housings are placed next to the tissue to help minimize artifacts due to patient motion during scanning. For interventional procedures, sterile, single-use biopsy grids serve to immobilize the breast and provide access windows for biopsy needle guidance.

7.3.4 System Components and Function

The following figures show the Tabletop's general set up for bilateral breast imaging, unilateral interventional and bilateral interventional procedures.

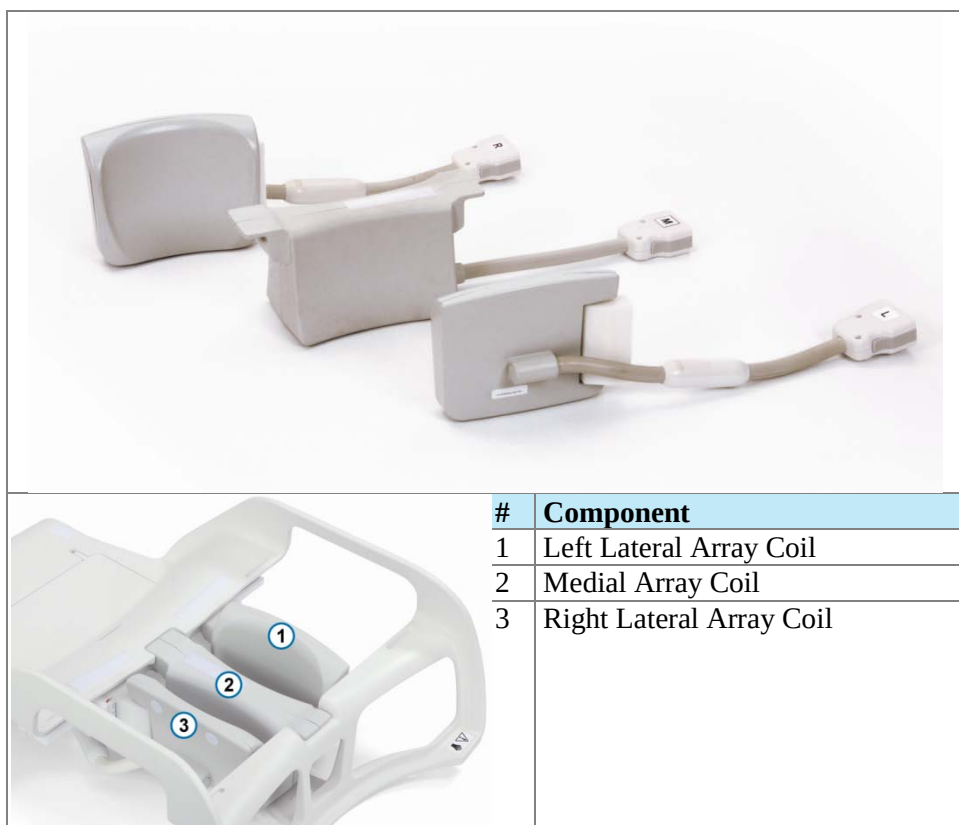


Figure 17 Bilateral Imaging Components and Setup (Without Padding)

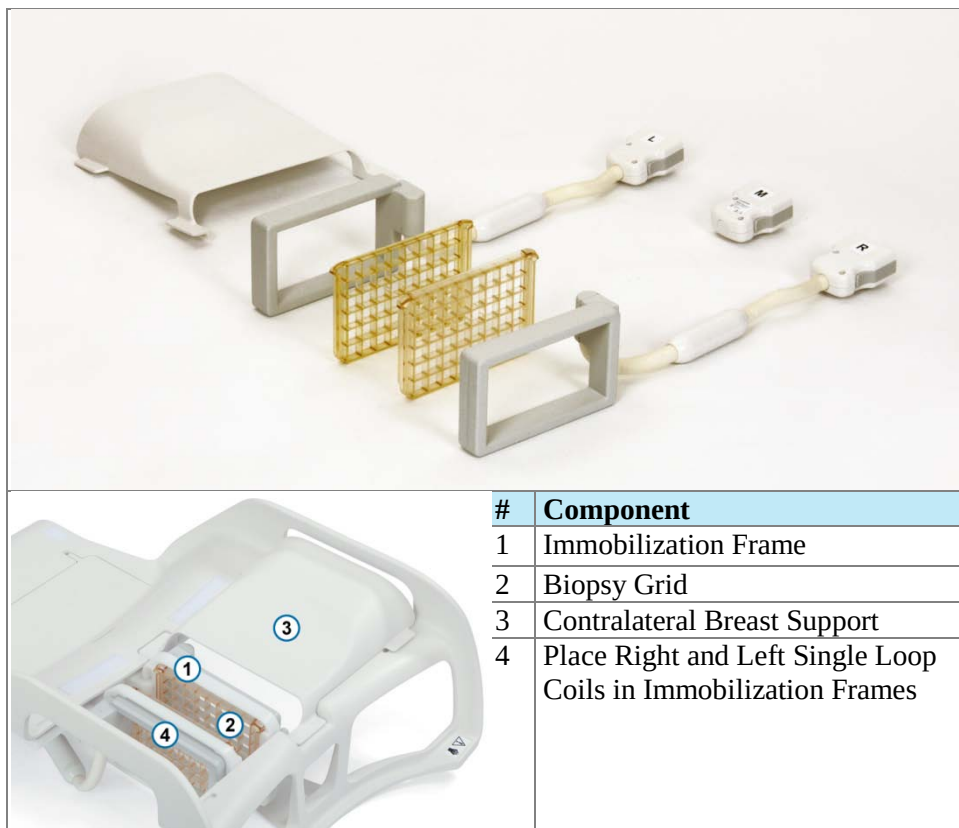


Figure 18 Unilateral Intervention Setup and Components (Without Padding)

NOTE: Biopsy grids are not included with the GE catalog offering. These products can only be ordered directly from Sentinelle Medical.

7.3.4.1. About the Contralateral Breast Support

The contralateral breast support holds the contralateral breast away to improve medial access during unilateral interventional procedures. The design of the breast support allows the breast of interest to fall further into the aperture for increased lateral posterior access to the breast during intervention.

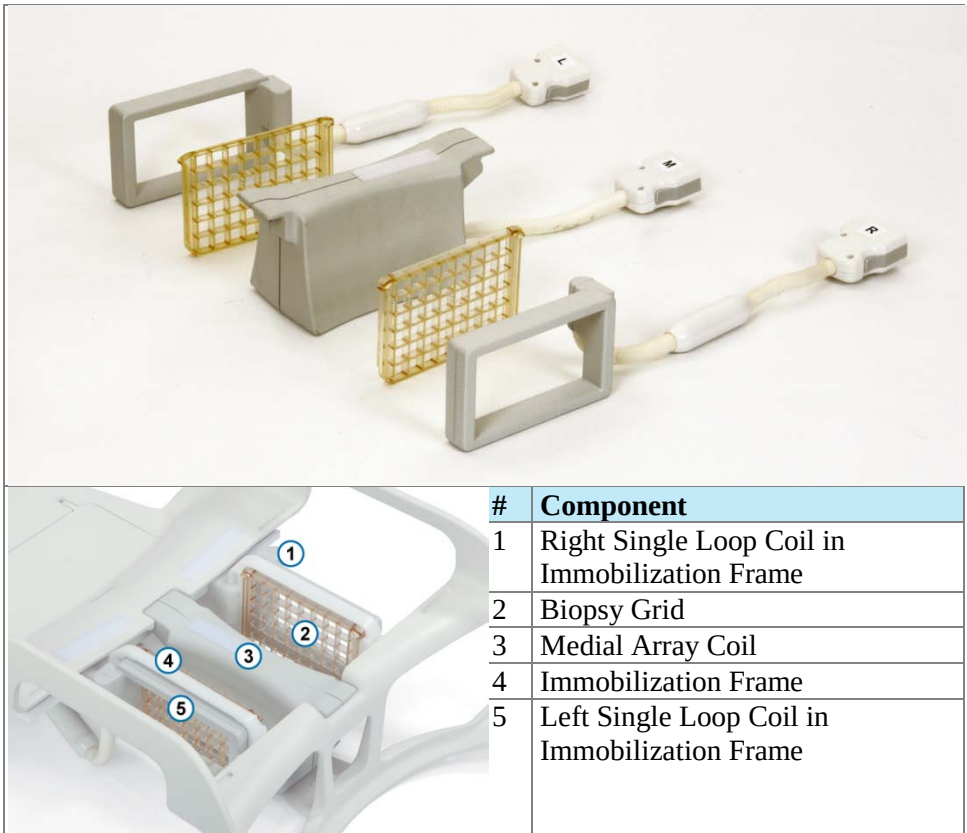


Figure 19 Bilateral Intervention Setup and Components (Without Padding)

7.3.4.2. Catchment Tray

For unilateral and bilateral interventional procedures the catchment tray must be in place. This is a safety mechanism ensuring the MR guided needle device can safely be directed into the breast tissue.



Figure 20: Inserting the Catchment Tray

7.3.4.3. Vanguard Cable



Plug in the Vanguard cable. Do not scan if the cable is unplugged or if there is an MR system warning. The Vanguard cable **must** be covered at all times during operation with the body pad (Figure 18).

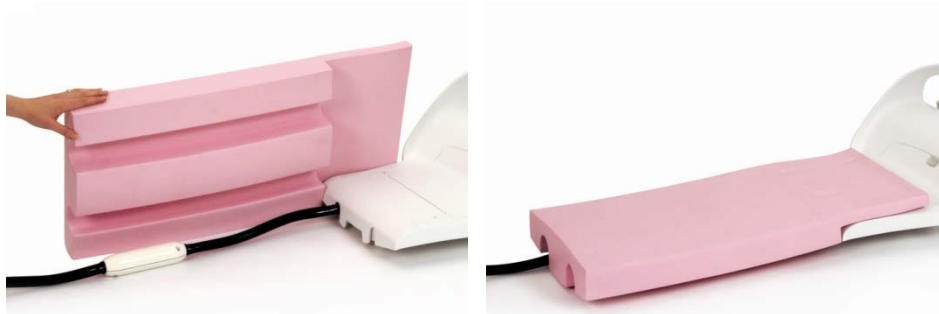


Figure 21 Covering the Tabletop Cable with the Body Pad

7.4 Coil Data Disk Installation Procedure

7.4.1 Personnel Requirements

Required Persons	Procedure	Finalization
1	15 mins	45 minutes

7.4.2 Tools and Test Equipment

Item	Qty	Part#
16Ch Breast Coils by Sentinelle Coil Definitions Disc	1	5407766
Phantoms and phantom positioners are needed for the MCQA test.	N/a	See sections 6.1 and 7.5

7.4.3 Required Conditions

Condition
The software build to install this coil must be 22.0 M4 or above

7.4.4 Procedure

NOTE: Do not plug in the coil and use AutoCoil to install the coil. This method will not install the coil protocol.

1. Make sure that the latest software build is loaded on the system prior to installing the Sentinelle coil. The Sentinelle protocol will only install if the required conditions are met.
2. Check to confirm if the 16Ch Sentinelle table/coil is already installed on the system software by opening the Service Browser under the Service Desktop Manager. In the Service Browser click on the Configuration tab, then the Coil Database Explorer under the configuration folder on the left.
3. Select "Click here to start this tool". The Coil DB Explorer window will open and under the "Coil Name" tab there is a "Clinical Coil Name" control. Click on this and there will be a dropdown list of all the logical coil names. There should be seven names that end with "SMI" (they will be listed alphabetically). This indicates the coil configs are present in the Coil DB. Reference Figure 24. If the coil is not present on the list proceed to step 6.

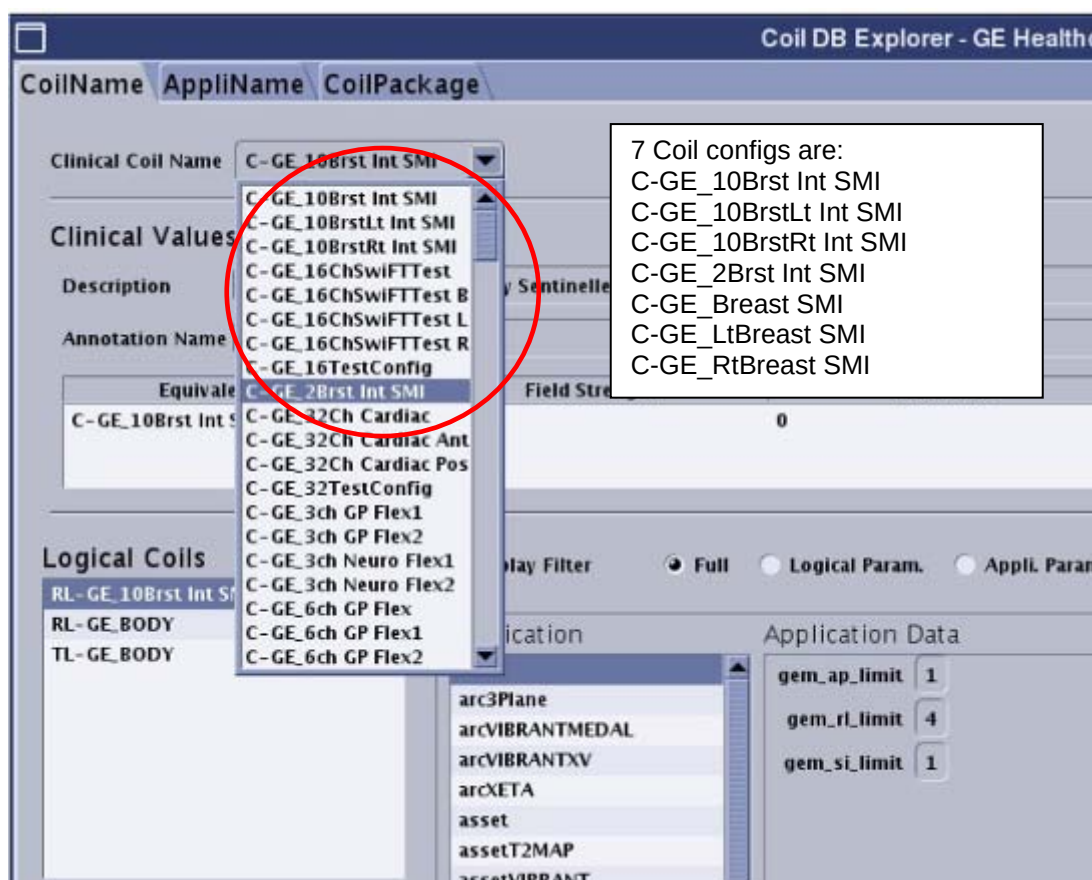


Figure 22: Confirmation of coil configurations installed

- The 16Ch Sentinelle table also has one protocol that must be on the system. Open the Protocol Management tab. Confirm the protocol installed by viewing the "Chest" protocols under the GE Protocol Library. The protocol titled "Breast Routine 16Ch Sentinelle" must be present as shown in Figure 25.

If protocol is not present and the coil CD was just installed one may have to exit the "Chest" protocols by selecting another area such as "Abdomen" and then reselecting "Chest" for the protocol to appear.



Figure 23: Confirmation of 16ch Sentinelle coil protocols installed

5. If the Sentinelle table configurations and protocols are on the system skip to the finalization steps in section 7.4.5.
6. To install the Sentinelle table configurations and protocols, insert the “16Ch Breast Coils by Sentinelle Coil Definitions Disc” into the default (typically upper) host DVD drive.
7. Open a C-Shell from the Service Desktop manager and type the following commands:


```
su <enter>
<root password>
mount /mnt/cdrom <enter>
/mnt/cdrom/autorun <enter>
```
8. A Coil Data Disk Install window will open. Type “i” to install the CD contents.
9. When the content of the CD installation is complete, the cursor will appear and you will see the message as shown in Figure 26. Press enter to exit the Coil CD installation window.

Coil Data Disk Install

```

Created //export/home/signa/tools/mc_qa_tool/3.0T_HD_Shoulder_Array_by_GE_AD_port.mctcfg
Created //export/home/signa/tools/mc_qa_tool/3.0T_MR750_HNS_by_GE_PosteriorHead_Face.mctcfg
Created //export/home/signa/tools/mc_qa_tool/3.0T_MR750_HNS_by_GE_PosteriorHead_Face.Unified.mctcfg
Created //export/home/signa/tools/mc_qa_tool/3.0T_MR750_HNS_by_GE_PostHeadChestHorseshoeTL.mctcfg
Created //export/home/signa/tools/mc_qa_tool/3.0T_MR750_HNS_by_GE_PostHeadChestHorseshoeTL.Unified.mctcfg
Created //export/home/signa/tools/mc_qa_tool/3.0T_MR750_HNS_by_GE_TL.mctcfg
Created //export/home/signa/tools/mc_qa_tool/3.0T_MR750_HNS_by_GE_TL.Unified.mctcfg
Created //export/home/signa/tools/mc_qa_tool/3.0T_DV_10Ch_Breast_Table_by_SMI.mctcfg
Created //export/home/signa/tools/mc_qa_tool/3.0T_DV_16Ch_Breast_Table_by_SMI.mctcfg
PASS
coil ID is not in DB
Installing protocols, if needed

chmod: cannot access '/usr/g/ufprotocols/.invalid/-$prot': No such file or directory
chmod: cannot access '/usr/g/ufprotocols/.invalid/-$prot': No such file or directory
Finished Execution

Installing protocols, if needed

chmod: cannot access '/usr/g/ufprotocols/.invalid/-$prot': No such file or directory
chmod: cannot access '/usr/g/ufprotocols/.invalid/-$prot': No such file or directory
Finished Execution

Installing protocols, if needed

chmod: cannot access '/usr/g/ufprotocols/.invalid/-$prot': No such file or directory
chmod: cannot access '/usr/g/ufprotocols/.invalid/-$prot': No such file or directory
Finished Execution

You must create a new saveINFO disc to make the changes permanent

Press 'enter' to exit

```

- Note the PASS message.

- The lines selected show that the Sentinelle packages installed from the CD for a 3T system.

- The statement that “coil ID is not in DB” is not an error message pertaining to the install.

Figure 24: CD Contents Installation Complete

- Exit the “Coil Data Disk Install” window and type <eject> in the original c shell. Remove the CD from the host DVD drive.
- To confirm the coil has been properly installed from the coil CD follow steps 2 to 4.

7.4.5 Finalization

- Perform a SaveInfo under Guided Install to make the changes permanent.
- Perform the Multi-Coil Quality Assurance (MCQA) test with this coil. Refer to Section 7.5.

GEHC ONLY	- Complete Product Locator information for serialized components. Failure do this may result in failure of your site to receive future FMI's. Instructions are below. - Go to GIB: http://gib.gehealthcare.com/gib/gib_MainMenu.jsp - After the log in screen, click on [Global Installed Base entries] - Under “Serial Changes to Installed Base”, enter the system ID of the system on which this coil is being installed. Click [Submit] - Click on [New Entries]
-----------	---

7.5 QA Scanning

Perform a QA scan of the phantoms to ensure all coils are working to specification. The QA scan must be completed for 16ch and 10ch configurations of the Vanguard.

*Note: QA scan test results should be determined using the procedure outlined below in **Test 1 and 2** and entered into the Quality Assurance SNR Section in the **Installation Verification Checklist** which can be found in the Appendix.*

Sentinel Medical recommends that the coils be checked for imaging integrity on a regular basis, by performing the Multi-Coil Quality Assurance (MCQA) procedure. The phantoms used for the QA scan are part of GE's System Phantom Set. The instructions below only apply to the Sentinel Vanguard system.

Note: If bubbles are apparent in the phantom, tap the outside of the phantom until these bubbles disappear.

Tools Required

The tools required to complete the MCQA procedures are outlined below in Test 1 and 2 are listed in the following table.

Description	GEHC Part #	SMI Part #	Qty
10cm Silicon Oil Spherical Phantom	2360034	-	2
Phantom positioner	5407765-11	SVC-00487	1

7.5.1 Test 1: 16-channel Bilateral Breast SNR Verification

1. Install Vanguard tabletop on MR scanner table.
2. Install and connect the three coil cables of the 16-channel coil set to the Vanguard table (refer to Figure 17) and plug in the Vanguard system cable into the P-port receptacle.
3. Place the phantom positioner on the medial coil as shown in Figure 25.



Figure 25: 16 Channel - Phantom Positioner Setup

4. Add spherical GE phantoms and adjust lateral coils so that lateral coils lightly touch the phantoms as shown in Figure 26.

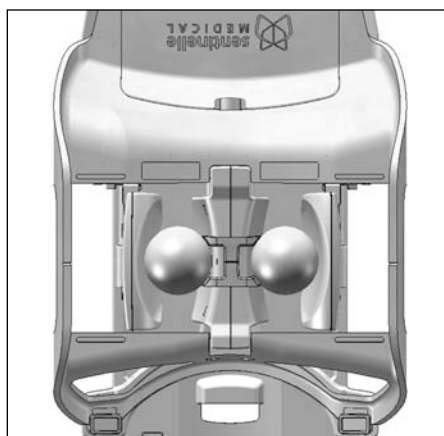


Figure 26: 16 Channel - Phantom Setup

5. Landmark using the notch on the edge of the patient support as shown in Figure 27.

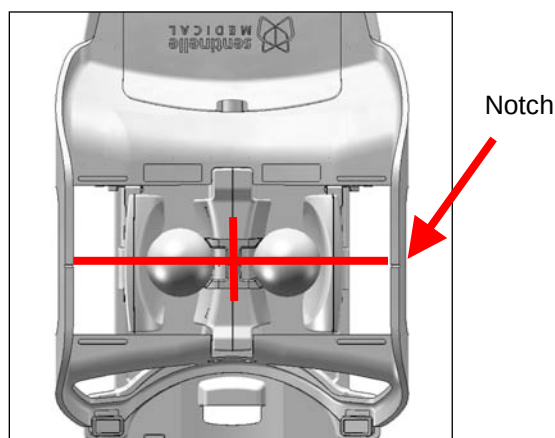


Figure 27: 16 Channel - Landmark Phantoms

6. Do not start a new patient; open the Tools box, and start MR Service Desktop. Select the top tab marked 'Image Quality.' In the side menu, select 'Multi-Coil QA Tool', then 'Click here to start this tool.' The program shown in Figure 28 will start:

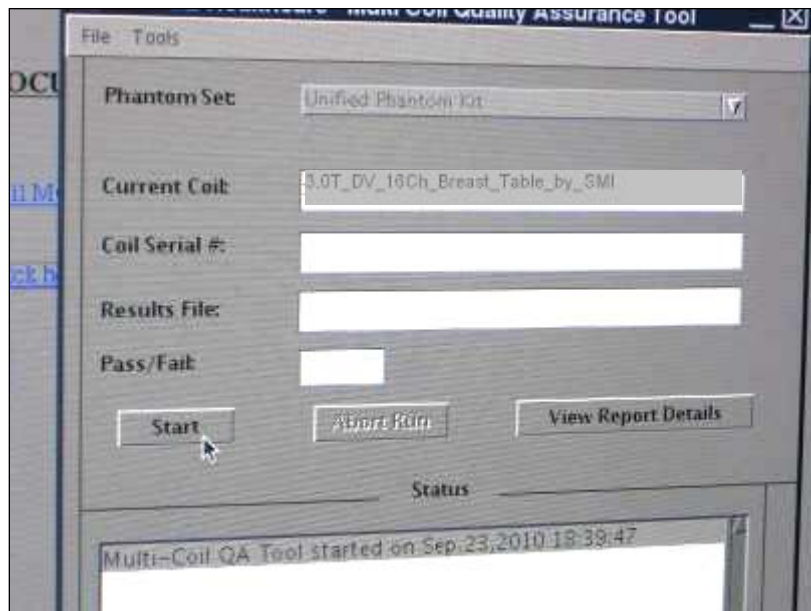


Figure 28: Screen Capture – MCQA Tool Start

7. Click 'Start', and answer 'Yes' to the dialog box that opens. The MCQA program will run for approximately 10 minutes, and test each channel of the system independently. When the program finishes, the screen capture shown in Figure 29 will display.

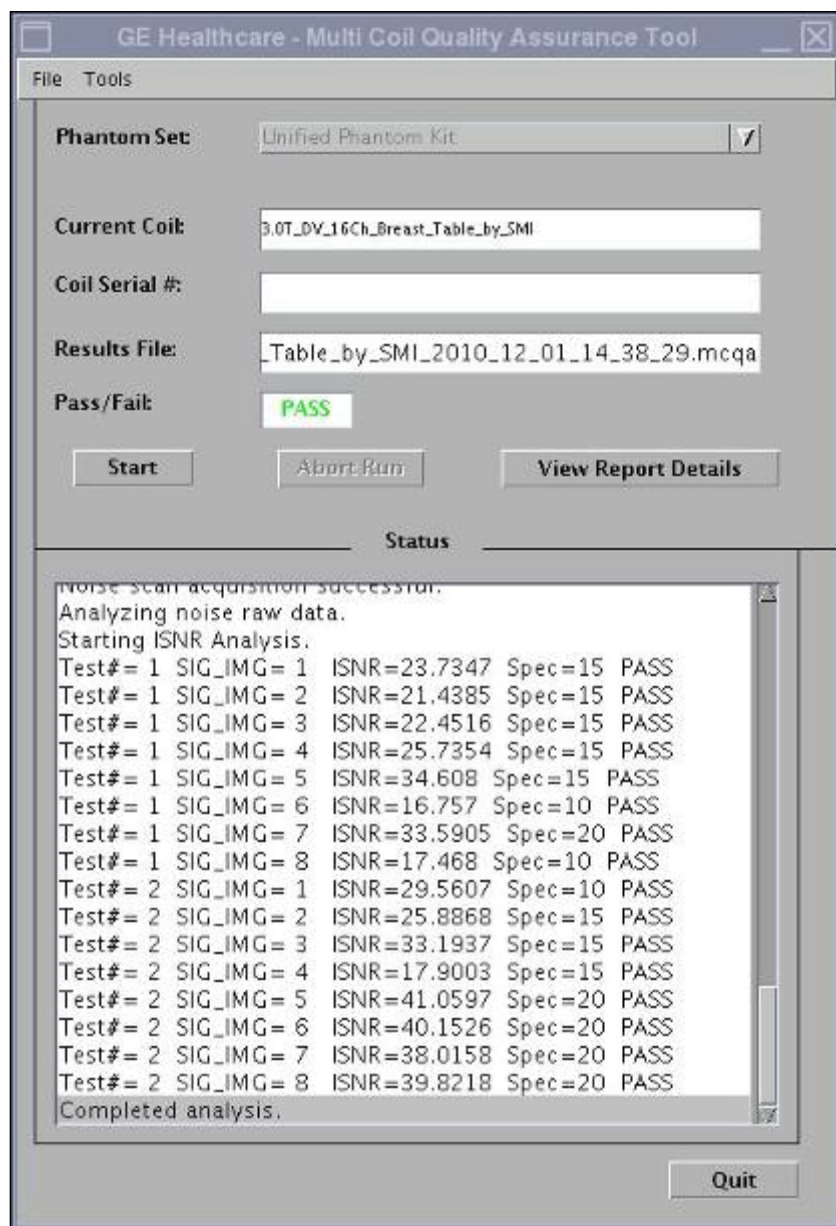


Figure 29: Screen Capture – MCQA Tool Finish

8. The MCQA program will display a PASS/FAIL indicator for each image (SIG_IMG in Figure 29) which corresponds to a particular channel on one of the coils tested. If you receive a FAIL indication, check that all coils are connected properly and rerun the test. If the test fails the second time refer to the table below.

MCQA Test Image	Associated Coil	GEHC P/N	SMI P/N
Test#=1 SIG_IMG=1	Left Lateral Array	5407765-5	4000428-11
Test#=1 SIG_IMG=2	Left Lateral Array	5407765-5	4000428-11
Test#=1 SIG_IMG=3	Left Lateral Array	5407765-5	4000428-11
Test#=1 SIG_IMG=4	Left Lateral Array	5407765-5	4000428-11
Test#=1 SIG_IMG=5	Medial Array	5407765-6	4000445-11
Test#=1 SIG_IMG=6	Medial Array	5407765-6	4000445-11
Test#=1 SIG_IMG=7	Medial Array	5407765-6	4000445-11
Test#=1 SIG_IMG=8	Medial Array	5407765-6	4000445-11
Test#=2 SIG_IMG=1	Medial Array	5407765-6	4000445-11
Test#=2 SIG_IMG=2	Medial Array	5407765-6	4000445-11
Test#=2 SIG_IMG=3	Medial Array	5407765-6	4000445-11
Test#=2 SIG_IMG=4	Medial Array	5407765-6	4000445-11
Test#=2 SIG_IMG=5	Right Lateral Array	5407765-4	4000429-11
Test#=2 SIG_IMG=6	Right Lateral Array	5407765-4	4000429-11
Test#=2 SIG_IMG=7	Right Lateral Array	5407765-4	4000429-11
Test#=2 SIG_IMG=8	Right Lateral Array	5407765-4	4000429-11

Additional information related to the specific channels used in each imaging configuration is available in section 8.1 *Imaging Troubleshooting*.

7.5.2 Test 2: 10-channel Bilateral Breast SNR Verification

1. Install Vanguard tabletop on MR scanner table.
2. Install and connect the three coil cables of the 10-channel coil set (refer to Figure 19) to the Vanguard table and plug in the Vanguard system cable into the P-port receptacle.
3. Locate phantom positioners on either side of the medial coil as shown in Figure 30.



Figure 30: 10 Channel - Phantom Positioner Setup

4. Add spherical GE phantoms and adjust lateral coils so that lateral coils lightly touch the phantoms as shown in Figure 31.



Figure 31: 10 Channel - Phantom Setup

5. Landmark using the notch on the edge of the patient support as shown in Figure 32.

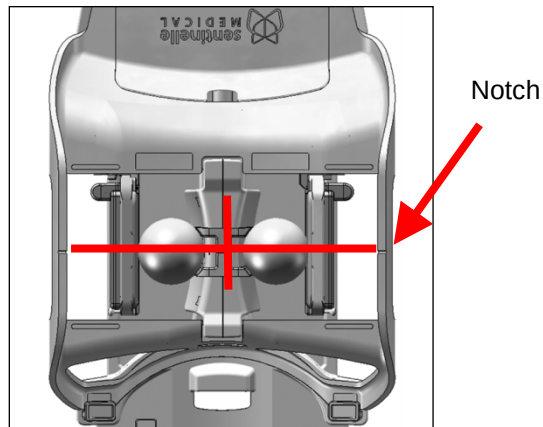


Figure 32: 10 Channel - Landmark Phantoms

6. Do not start a new patient; open the Tools box, and start MR Service Desktop. Select the top tab marked 'Image Quality.' In the side menu, select 'Multi-Coil QA Tool', then 'Click here to start this tool.' The program shown in Figure 33 will start:

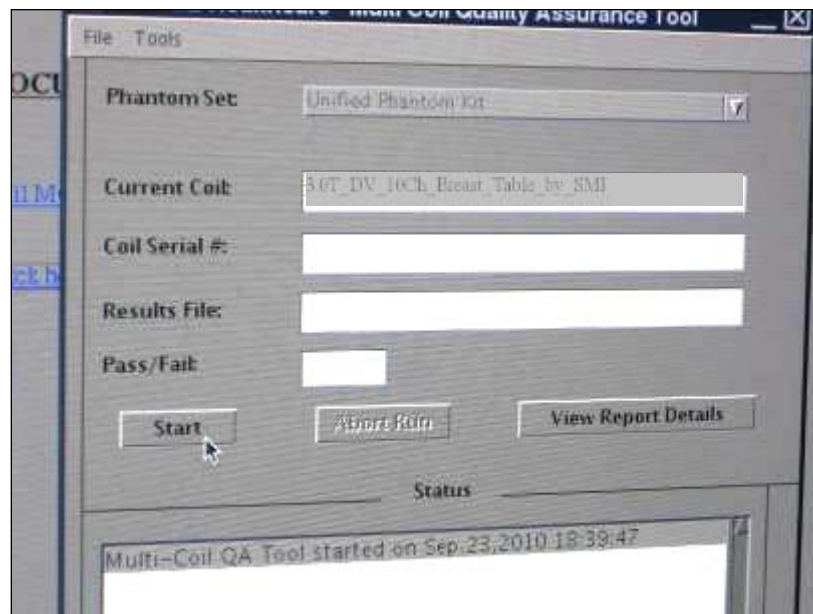


Figure 33: Screen Capture – MCQA Tool Start

7. Click 'Start', and answer 'Yes' to the dialog box that opens. The MCQA program will run for approximately 10 minutes, and test each channel of the system independently. When the program finishes, the screen capture shown in Figure 34 will display.

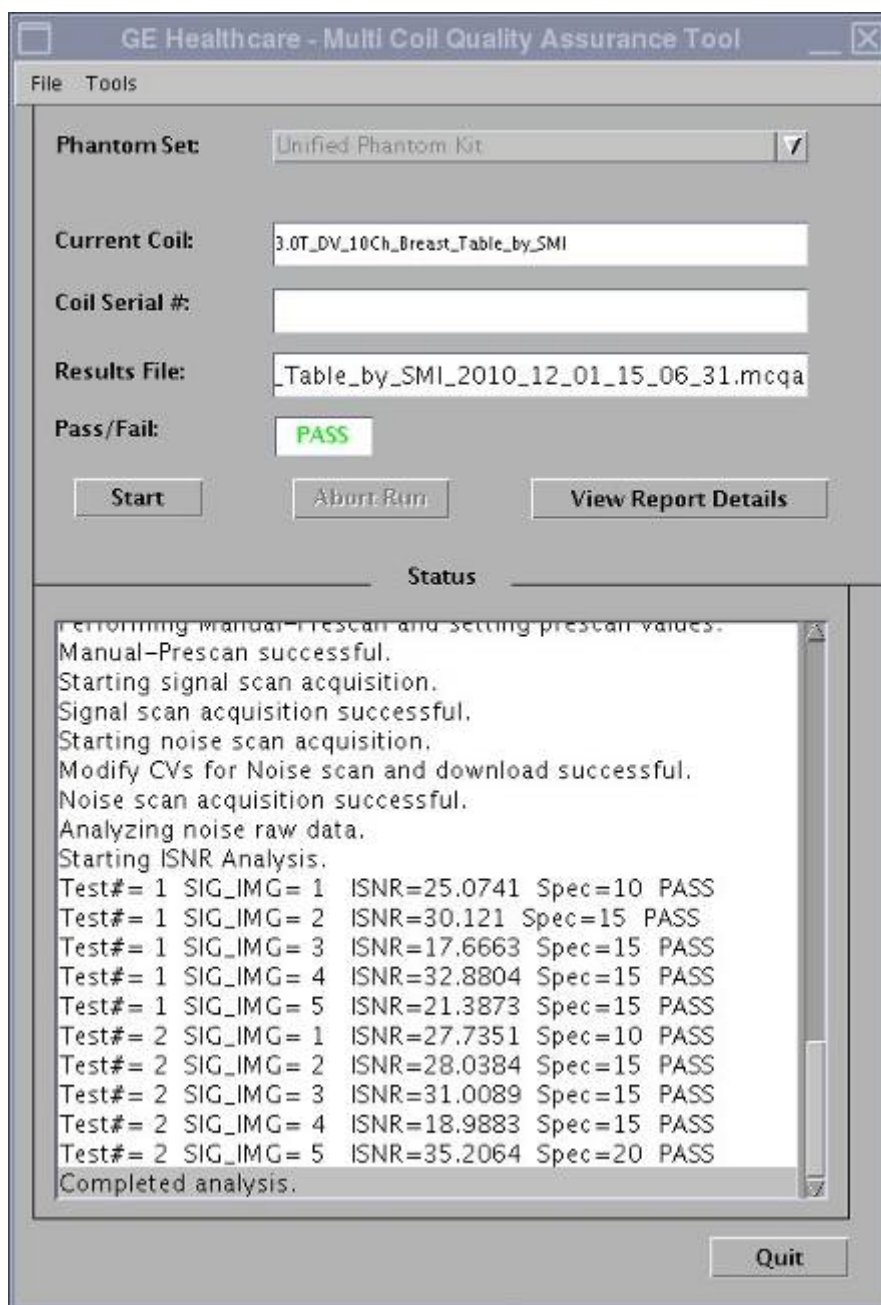


Figure 34: Screen Capture – MCQA Tool Finish

8. The MCQA program will display a PASS/FAIL indicator for each image (SIG_IMG in Figure 34) which corresponds to a particular channel on one of the coils tested. If you receive a FAIL indication, check that all coils are connected properly and rerun the test. If the test fails the second time refer to the table below.

MCQA Test Image	Associated Coil	GEHC P/N	SMI P/N
Test#=1 SIG_IMG=1	Left Single Loop	5407765-2	4000452-11
Test#=1 SIG_IMG=2	Medial Array	5407765-6	4000445-11
Test#=1 SIG_IMG=3	Medial Array	5407765-6	4000445-11
Test#=1 SIG_IMG=4	Medial Array	5407765-6	4000445-11
Test#=1 SIG_IMG=5	Medial Array	5407765-6	4000445-11
Test#=2 SIG_IMG=1	Medial Array	5407765-6	4000445-11
Test#=2 SIG_IMG=2	Medial Array	5407765-6	4000445-11
Test#=2 SIG_IMG=3	Medial Array	5407765-6	4000445-11
Test#=2 SIG_IMG=4	Medial Array	5407765-6	4000445-11
Test#=2 SIG_IMG=5	Right Single Loop	5407765-3	4000448-11

Additional information related to the specific channels used in each imaging configuration is available in section 8.1 Imaging Troubleshooting.

7.5.3 Troubleshooting tips

1. Ensure that no patient has been started before beginning the MCQA procedure.
2. Landmark the coils before starting the MCQA tool.
3. Only use the spherical GE phantoms (GE Model 2360034), and the Sentinelle phantom positioners.
4. Ensure that all coils are touching the phantoms.
5. When the MCQA tools starts, you may see the message shown in Figure 35. If so, click 'Yes'.



Figure 35: MCQA Warning Message

7.6 Final Verification Checklist

Final verification of install is an important series of functional and visual checks which are undertaken to ensure that the installation procedure was executed correctly.

SMI ONLY	Complete the Installation Verification Checklist (section 14 Appendix A) and submit to SMI Service department.
GEHC ONLY	Complete the Installation Verification Checklist (section 14 Appendix A) and store on site.

7.7 System Storage

When installation is complete, store the system as directed by the customer or inside the scanner room in an area where it will not pose a hazard or impede access to any part of the room. The storage environment should be **1.5m x 1m x 1m** or **60in x 40in x 40in** and must meet the criteria listed in the table below. The weight of the system is 9kg or 20lb.

	Minimum	Maximum
Ambient Temperature (°C)	-30	+65
Relative Humidity (non-condensing)	10%	95%
Atmospheric Pressure	500 hPa	1060 hPa

8 Troubleshooting

GE FE's shall service systems that were sold by GEHC. Coils sold by GEHC will have a "Distributed by GE Healthcare" label next to the rating plate on the underside of the patient support.

If the Vanguard is not functioning correctly, steps must be taken to troubleshoot the problem by analysing the symptoms displayed and identifying the cause.

The two classifications of problems which may occur are **imaging** or **mechanical**.

8.1 Imaging Troubleshooting

Imaging problems may occur due to damage or malfunction of any component of the RF system, or incorrect use of the system as a whole. Imaging problems can be addressed by analysing and modifying the protocols utilised on the scanner or replacing damaged RF components. Initially consult the following table to troubleshoot any imaging problems.

Problem	Possible Solution(s)
You are unable to pre-scan or are scanning and receiving no signal.	• Verify that you have selected the correct coil configuration. Refer to 8.3 <i>Coil Configurations</i> to verify the proper coils have been selected. • Verify that the coils are: - o Properly attached to the immobilization frames. - o Properly connected to the signal lines. - o Correspond to the coil configuration. • Verify that the landmark is correct • Verify that the Tabletop cable is properly connected to the MRI. • Verify that the scan locations and any FOV offsets are correct Note: When selecting two separate 2D imaging volumes, the scan may select the middle slice of the entire set of slices for prescan. Ensure the middle slice is covering an appropriate volume of interest. If a scan still cannot be performed, remove all of the coils, terminate all of the signal lines, and attempt to scan using the body coil and the identical imaging sequence and parameters. If there is still no signal received, the problem is probably with the MRI system. Please contact your service representative.
Signal quality obtained in the quality assessment check is not acceptable or clinical imaging scan quality is not acceptable.	• Ensure the protocol used in the periodic quality assessment check is identical to initial baseline scan. • Verify that all coils are properly selected. • Verify that coils are in proper positions. • Verify that there are no loops in the cables. • Verify that the imaging area is free from ferromagnetic and foreign objects. • Verify that the center frequency is within the adjustment range of your system. • Ensure electrical contacts on the coils and the MRI connectors are clean and free from obstruction. If they are not clean, they can be cleaned with a cotton swab and isopropyl alcohol.
A dark region appears in the image.	• Verify there are no ferromagnetic or foreign objects within the imaging volume.

If the problem can not be corrected, inspect the housings, cables and connectors of all RF components for damage. If an RF component is replaced, complete the *QA Scanning* according to Section 7.5.

8.2 Mechanical Troubleshooting

Mechanical problems can be addressed by identifying and replacing damaged or worn components or FRUs. *Section 11 Replacement Procedures* provides guidance for components that are not replaceable by the user.

8.3 Coil Configurations

Imaging Configuration	Connect to Left Cable Plug	Connect to Middle Receptacle	Connect to Right Cable Plug	Configuration to Select on Scanner	Channels Used by Coils
16ch Bilateral	Lateral Array Coil Left	Medial Array Coil	Lateral Array Coil Right	Breast SMI	Left: 1 – 4 Medial: 5 – 12 Right: 13 – 16
8ch Left Breast Unilateral	Lateral Array Coil Left	Medial Array Coil	Lateral Array Coil Right	LtBreast SMI	Left: 1 – 4 Medial: 5 – 8 Right: Inactive
8ch Right Breast Unilateral	Lateral Array Coil Left	Medial Array Coil	Lateral Array Coil Right	RtBreast SMI	Left: Inactive Medial: 9 – 12 Right: 13 – 16
10ch Interventional Bilateral	Single Loop Coil Left	Medial Array Coil	Single Loop Coil Right	10Brst Int SMI	Left: 3 Medial: 5 – 12 Right: 13
5ch Interventional Unilateral Left	Lateral Single Loop Coil Left	Medial Array Coil	Lateral Single Loop Coil Right	10BrstLt Int SMI	Left: 3 Medial: 5 – 8 Right: Inactive
5ch Interventional Unilateral Right	Lateral Array Coil Left	Medial Array Coil	Lateral Single Loop Coil Right	10BrstRt Int SMI	Left: Inactive Medial: 9 – 12 Right: 13
2ch Unilateral Breast	Lateral Single Loop Coil Left	Medial Array Coil	Lateral Single Loop Coil Right	2Brst Int SMI	Left: 3 Medial: Inactive Right: 13

9 FRU List with GE Part Numbers

FRUs are parts or assemblies that can be replaced if damaged. The following tables list all FRUs that are available for ordering from Sentinelle or GEHC.

9.1 Patient Support FRUs

GEHC Part #	Sentinelle Part #	Part Description	Section #
5408232	4000189-11	Patient Support Assembly	6.1.1
5408233	SVC-00330	Contralateral Support	6.1.1
5408234	SVC-00331	Catchment	6.1.1
5408235	SVC-00319	Shoulder Bridge	6.1.1
5408236	SVC-00318	Headrest Assembly	6.1.1
5408237	SVC-00337	Headrest Mirror Assembly	-
5408240	4000210-11	Compression Slider Left	6.1.1
5408241	4000290-11	Compression Slider Right	6.1.1
5408239	SVC-00307	Compression Plate	6.1.1
5408244	4000197-11	Compression Frame Ass'y Right	6.1.1
5408245	4000196-11	Compression Frame Ass'y Left	6.1.1
5408251	SVC-00347	Compression Rail	10.1, 11.1
5408250	SVC-00346	Table Alignment Tab	10.1, 11.2
5408248	SVC-00344	Patient Support Cover	10.1, 11.4
Contained in 5408246	1659A2	Patient Support Cover Catch	10.1, 11.5
5407765-11	SVC-00487	GE Spherical Phantom Positioner	6.1.1

9.2 RF System FRUs

GEHC Part #	Sentinelle Part #	Part Description	Section #
5407765-2	4000452-11	Single Loop Left 3T16Ch	6.1.2
5407765-3	4000448-11	Single Loop Right 3T16Ch	6.1.2
5407765-5	4000428-11	Lateral Array Left 3T16Ch	6.1.2
5407765-4	4000429-11	Lateral Array Right 3T16Ch	6.1.2
5407765-6	4000445-11	Medial Array 3T16Ch	6.1.2
5407765-7	4000349-11	Medial Plug 3T16Ch	6.1.2
5407765-8	4000444-11	Interface Box 3T16Ch	6.1.2, 11.3

9.3 Padding Kit FRUs

GEHC ONLY	Pads and straps are designed as customer consumable components. They are not intended to be covered by any GE service contract. If pads and straps are replaced they should be billed to the customer.
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GEHC Part #	Sentinelle Part #	Part Description	Section #
5408291	SVC-00325	Pad Arm Support	6.1.3
5408289	SVC-00328	Body Pad	6.1.3
5408288	SVC-00309	Wedge Pad	6.1.3
5408287	SVC-00308	Headrest Pad	6.1.3
5408290	4001152-51	Padding Kit Table	-
Contained in 5408290 Kit	SVC-00322	Medial Pad	6.1.3
Contained in 5408290 Kit	SVC-00320	Foot Pad	6.1.3
Contained in 5408290 Kit	SVC-00326	Ramp Pad	6.1.3
Contained in 5408290 Kit	SVC-00327	Shoulder Bridge Pad	6.1.3
E8811EB	46-25845G1	Kit/Strap	6.1.3
5408286	4001129-51	Asset Pad Kit (2 Lateral and 1 Medial)	-

9.4 Riser Kit FRUs

GEHC Part #	Sentinelle Part #	Part Description	Section #
5407765-10	ASY-04625	Riser	6.1.5
5407765-13	SVC-00486	Riser Body Pad	6.1.5
5407765-14	SVC-00485	Riser Front Pad	6.1.5

10 FRU and Fastener Locations

10.1 Patient Support FRUs

Item	GEHC FRU #	SMI FRU #	FRU Description	Fasteners	
				Type	Qty
(A)	5408248	SVC-00344	Patient Support Cover	-	-
(B)	Contained in 5408246 Tabletop Spare Parts Kit	Contained in 4001157-11	Patient Support Cover Catch	(2) PHMS Slot 4-40x0.25L Brass	8
(C)	-	-	Not a FRU – removal required for Patient Support Cover replacement	(1) BHCS 6-32x0.25L Brass	8
(D)	5408251	SVC-00347	Compression Rail	(3) BHCS 10-32x0.75L	8
(E)	5408250	SVC-00346	Table Alignment Tab	(1) BHCS 6-32x0.25L Brass	8
(F)	5408232	4000189-11	Patient Support Assembly	-	-

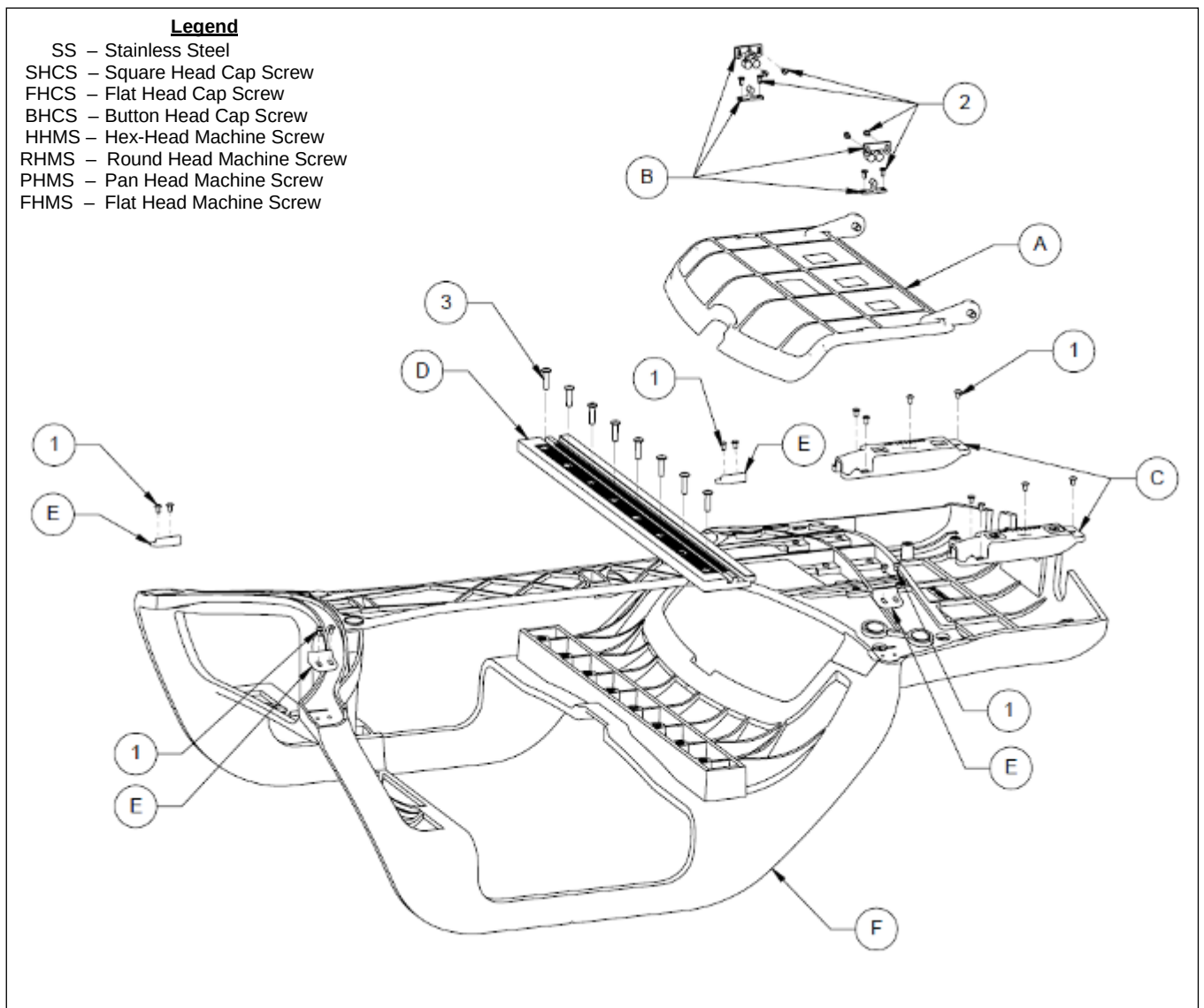


Figure 36

11 Replacement Procedures



WARNING

If any BBPs are present, stop work and notify customer that the Vanguard must be cleaned prior to performing service.

Most components are direct replacements requiring no tools or training to perform. Replacement procedures are defined for components that should be cleaned, replaced or repaired by the field service technician and require testing after replacement.

11.1 Compression Rail

SMI Part #: SVC-00347
GEHC Part #: 5408251
Time required: 20 minutes
Tools required: 1/8" Allen wrench
Personnel required: 1

1. Place Patient Support upside down on a soft/protected surface to access Compression Rail.
2. Take note of orientation of Compression Rail for correct replacement.
3. With a 1/8" Allen wrench, remove 8 screws which fasten Compression Rail.
4. Replace Compression Rail and replace 8 screws.

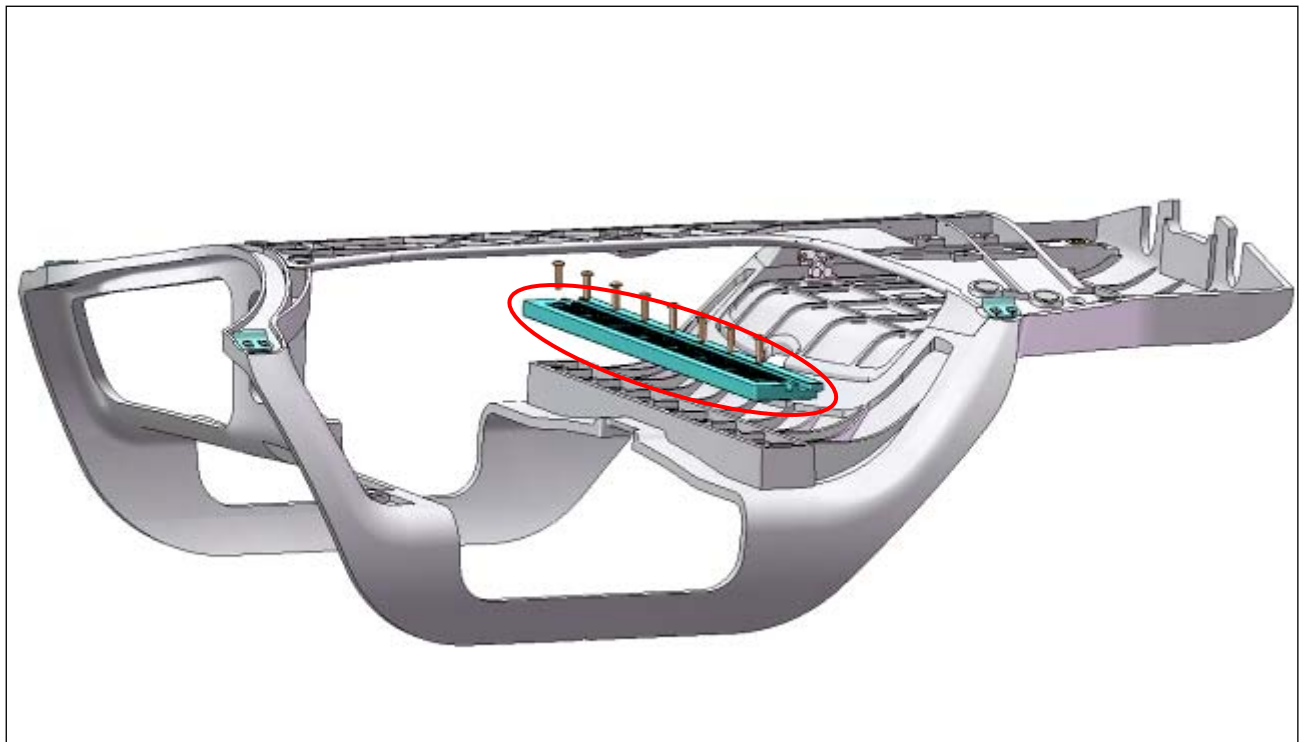


Figure 37

Finalization Steps

- i. Test function of sliders on Compression Rail by sliding and locking them on several positions along the length of the Compression Rail.

11.2 Table Alignment Tab

SMI Part #: SVC-00346
GEHC Part #: 5408250
Time required: 20 minutes
Tools required: 5/64" Allen wrench
Personnel required: 1

1. Place Patient Support upside down on a soft/protected surface to access Table Alignment Tab(s).
2. Note position of alignment tab by marking a reference line with a delible writing utensil on the Patient Support.
3. Unscrew and remove the two round head cap screws on alignment tab(s) and remove alignment tab.
4. Place new alignment tab on patient support carefully aligning it to reference line marked in step 2.
5. Tighten round head screws while ensuring tab does not shift.

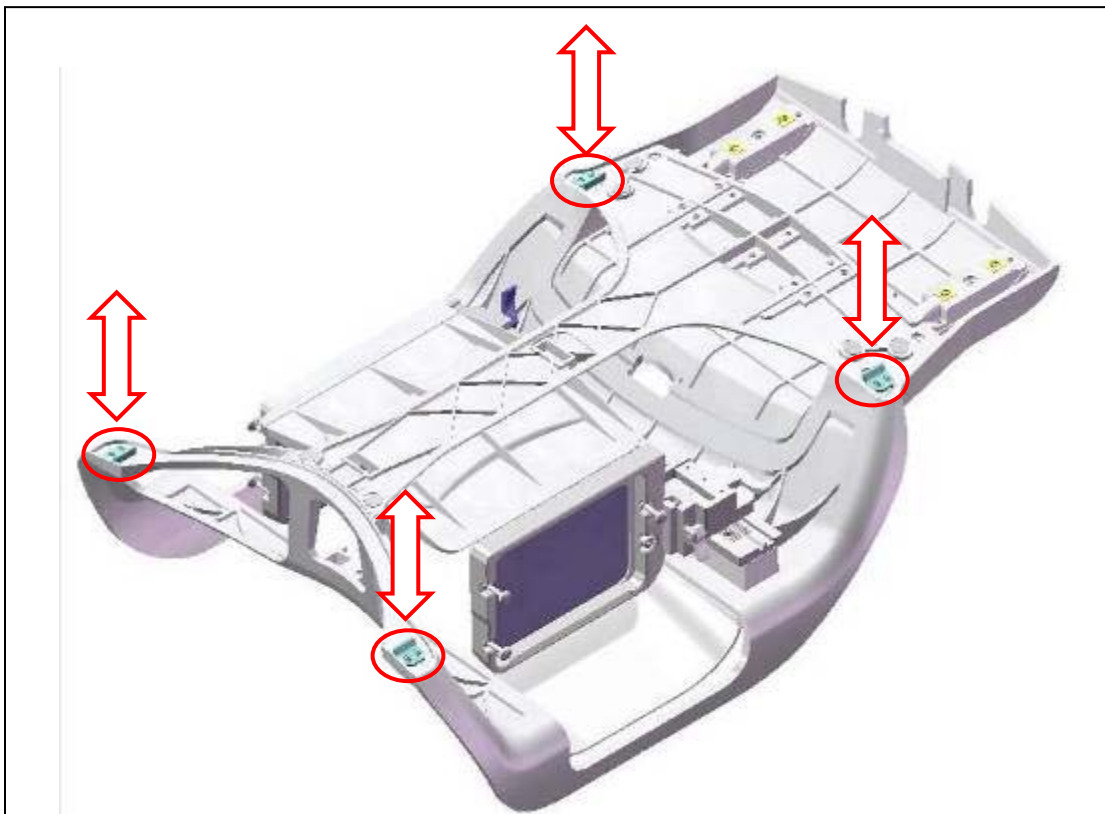


Figure 38

Finalization Steps

- i. Place Patient Support on GE table and ensure that it fits flatly on scanner bed and remains stationary.

11.3 Interface Box

SMI Part #: 4000444-11
GEHC Part #: 5407765-8
Time required: 10 minutes
Tools required: none
Personnel required: 1

1. Place Patient Support upside down on a soft/protected surface to access Interface Box.
2. Turn each fastener $\frac{1}{4}$ turn to loosen.
3. Remove Interface Box.
4. Mount replacement interface box and turn each fastener $\frac{1}{4}$ turn to tighten.

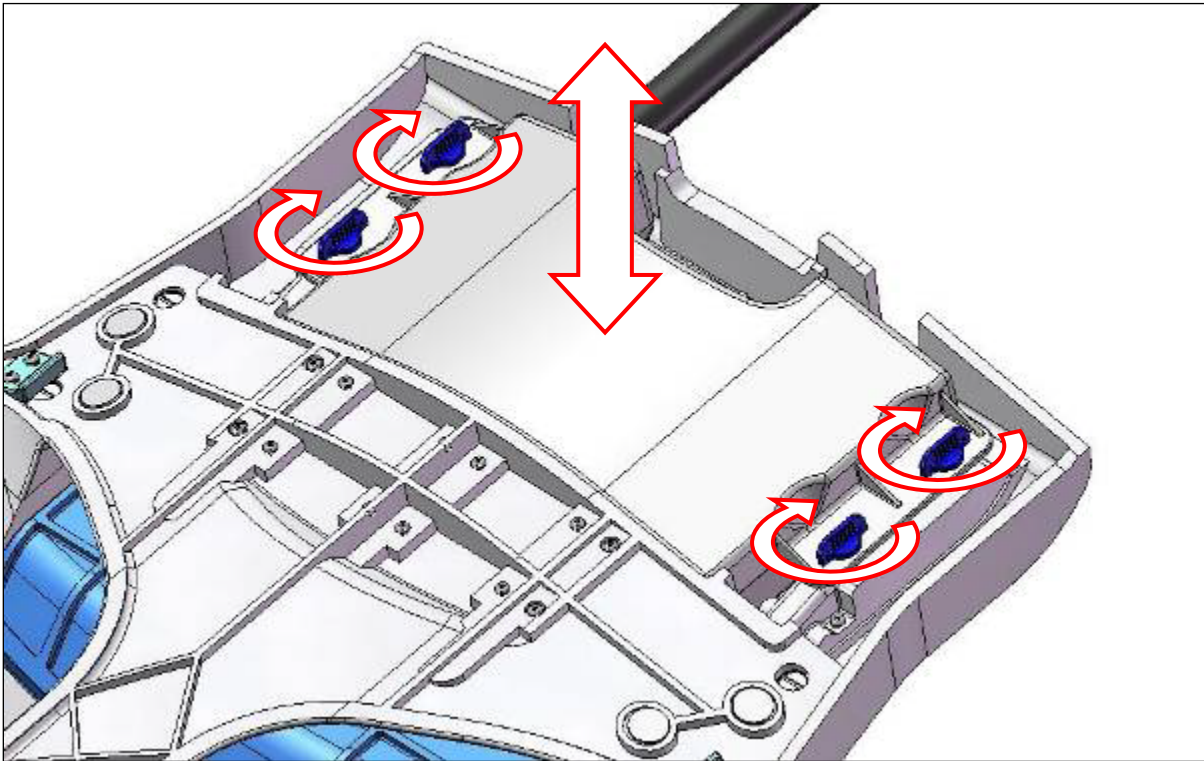


Figure 39

Finalization Steps

- i. Perform QA test in 7.5 QA *Scanning* section and record results for service report.

11.4 Patient Support Cover

SMI Part #: SVC-00344
GEHC Part #: 5408248
Time required: 20 minutes
Tools required: 5/64" Allen wrench
Personnel required: 1

1. Place Patient Support upside down on a soft/protected surface.
2. Remove Interface Box as detailed in section *11.3 Interface Box*.
3. Remove four screws which fasten each cover mounting bracket using a 5/64" Allen wrench.
4. Slide pins (shown in green) out of mounting holes in cover.
5. Remove damaged Patient Support Cover and replace with new Patient Support Cover
6. Insert hinge pins.
7. Re-attach mounting brackets.
8. Replace Interface Box.

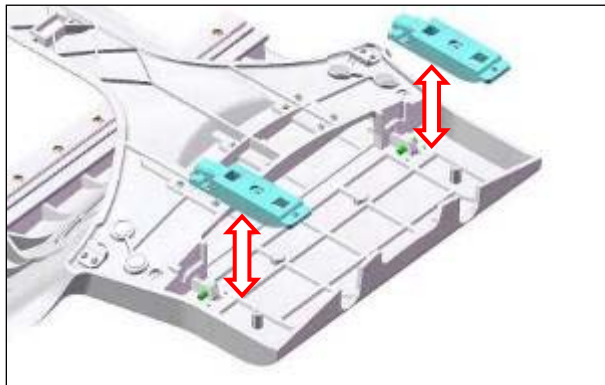


Figure 40



Figure 41

Finalization Steps

- i. Test function of new Patient Support Cover by opening and closing it.
- ii. Ensure no interference, misalignment or noises are evident.

11.5 Patient Support Cover Catch

SMI Part #: 1659A2
GEHC Part #: Contained in 5408246
Time required: 20 minutes
Tools required: Slot Screwdriver
Personnel required: 1

1. Open the Patient Support Cover.
2. Remove the striker from the underside the Patient Support Cover by unfastening two mounting screws.
3. Remove the rotary catch from the Patient Support by unfastening two mounting screws.
4. Remove the rotary catch and striker and replace with a new rotary catch and striker.
5. Partially fasten the rotary catch to the Patient Support with two screws.
6. Fasten the striker to the underside the Patient Support Cover with two screws.
7. With the cover closed, position and hold in place the catch assembly. Open the cover and tighten the screws.

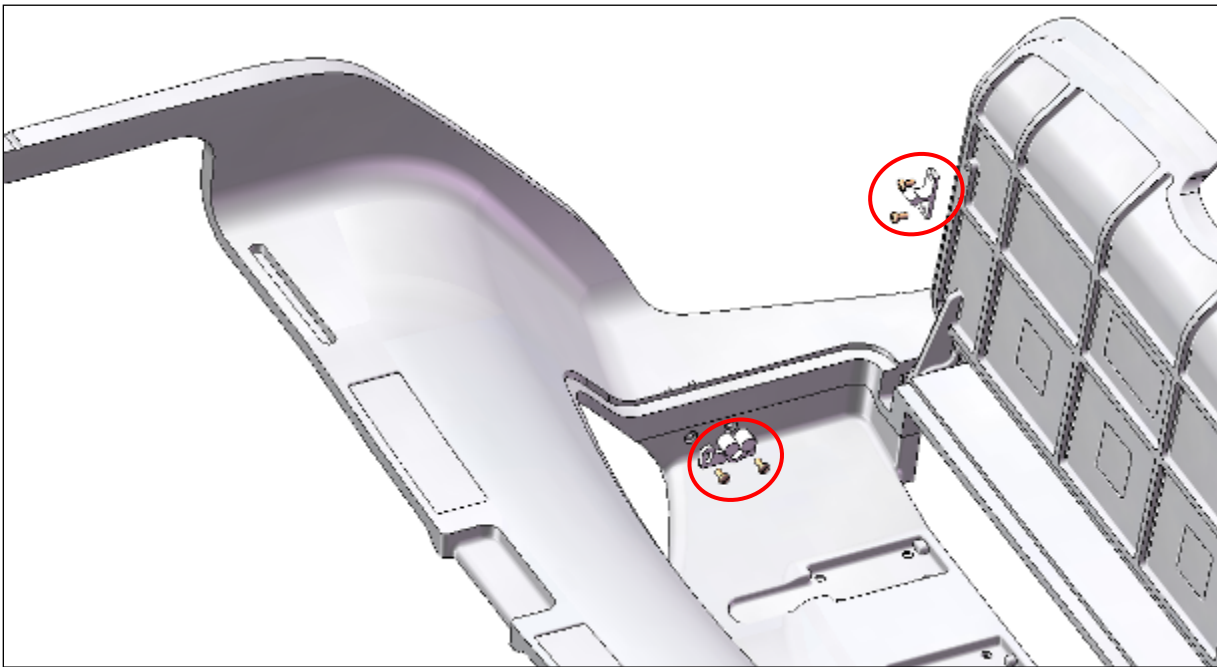


Figure 42

Finalization Steps

- i. Test function of new Patient Support Cover Catches by opening and closing the Patient Support Cover.
- ii. Ensure no interference, misalignment or noises are evident.
- iii. Inspect the rotary catches for any scratching or roughness which would indicate striker misalignment.

12 Appendix

Vanguard Tabletop for GE MR750 and MR750w Installation Verification Checklist

Install Date:	<u> D D M M M 20 Y Y </u>	Location:	<u> CITY, STATE / PROVINCE , _ _ _ </u>
Customer Name:	<u> </u>		
Vanguard Serial #:	<u> _ _ _ _ _ _ </u>	Configuration:	16 Channel
Scanner:	3.0T GE Discovery	Software Rev:	2 <u> _ _ _ _ _ _ </u>

Installation

- ☐ All components unpackaged, inspected and free of blemishes or damage.
- ☐ Patient support fits flatly on scanner bed and remains stationary.
- ☐ Patient support is centered and does not contact bore or any other surface.
- ☐ System is clean and free of fingerprints or debris.
- ☐ All coils snap in and hold firmly in the compression frames.
- ☐ Compression sliders slide smoothly and locks (2) work without excessive force.
- ☐ All padding is present, free of rips or damage and installed as per reference card.

SNR/QA Verification

- ☐ 16ch BL MCQA Result: Pass / Fail (Circle Result)
- ☐ 10ch BL MCQA Result: Pass / Fail (Circle Result)
- ☐ SNR protocol is saved on scanner console.
- ☐ SNR results for all configurations are recorded in the user manual.

SMI ONLY	☐ Any results not in specification reported immediately to SMI Service – 1-866-735-3744.
	☐ For any failing results, DICOM data saved to USB/DVD storage.

Pre-Departure

- ☐ All parts and accessories are present and shortages reported immediately.
- ☐ Accessories are neatly grouped for storage and/or as directed by customer.

SMI ONLY	☐ Fill out calling card and leave it in a visible location.
	☐ Any items not checked must be reported immediately by phone to SMI Service Department.
	☐ After completing this checklist, please provide a signed copy to the SMI Service Department.
GEHC ONLY	☐ After completing this checklist, please store a signed copy on site.

All items checked confirms Vanguard installation successfully completed.

Name: <u> </u>	Position: <u> </u>	Company: <u> </u>
Signed: <u> </u>	Date: <u> D D M M M 20 Y Y </u>	
Notes: <u> </u>		
<u> </u>		
<u> </u>		